

Logarithmic Communication for Low-Degree Slices and Product Domains

with Spectral–Lattice Stability

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Abstract

We study the adjacent two-value cross-intersection problem that occurs in an equivalent formulation of the Log-Rank Conjecture. We prove a spectral–lattice rectangle theorem for an arbitrary sparse right family: a covariance gap away from a real defect space controls the Hamming distance of every opposite row from a Boolean template, while low covariance on an integer generating set yields a large joint atom by a discrete entropy argument. The resulting rectangle bound depends on an exact Boolean tube radius and on the entropy of integer generator statistics, but has no exponential dependence on the dimension of the defect space or on its number of Boolean points.

We also prove an atomic product transfer theorem. If every Boolean degree-one function on each factor is an atom or its complement, and atom families have polynomial growth in centered rank with exponent α , then every row-degree-one Boolean matrix on the Cartesian product has deterministic communication complexity at most $\alpha \log_2(\text{rank } M + 1) + O(1)$. Conversely, a degree-one selector partition embeds arbitrary Boolean matrices, giving an abstract tractable/universal dichotomy.

In a different direction, let a Boolean matrix have the complete Johnson slice $\binom{[n]}{k}$ as its column domain, and suppose every row has slice degree at most s . Combining slice query-complexity bounds with rank pooling and a new symmetric-tree canonicalization, we prove

$$D(M) = O(s^8 \log(\text{rank } M + 1) + s^{16})$$

whenever $\min(k, n - k) \geq 144s^4$. Thus the Log-Rank Conjecture holds throughout this class even when $s = \text{polylog}(\text{rank } M)$. The same bound holds on the full cube without a layer-depth hypothesis, with a sharper dependence on s ; in the deeper regime $\min(k, n - k) \geq C^s$, we also obtain the sharper bound $O(s^3(\log(\text{rank } M + 1) + s))$. At the opposite boundary $k \leq s$, every Boolean function on the layer already has degree at most s , so arbitrary Boolean matrices occur. Qualitatively, the deep-layer hypothesis can be replaced by the sharp junta threshold $\min(k, n - k) \geq 2s$: for every fixed s , we obtain $D(M) = O_s(\log(\text{rank } M + 1))$. More generally, if every row on a symmetric or complete-product coordinate domain has bounded-answer query depth q , then

$$D(M) \leq q(\lceil \log_2(\text{rank } M + 1) \rceil + (q + 1)\lceil \log_2 Q \rceil),$$

where Q bounds the number of answers to a query. The same bounded-answer transfer proves $O_Q(s^8 \log(\text{rank } M + 1) + s^{16})$ communication for degree- s rows on complete product alphabets of size at most Q , and on Q -color multislices whose minimum color multiplicity is at least $216s^4$. An unsolvent-restriction theorem extends all these protocols to sparse column subsets. In particular, a uniform random $O(\binom{n}{s} \log \binom{n}{s})$ -column subfamily of a Johnson layer is, with high probability, simultaneously sufficient for every degree- s Boolean row family. Projection-DPP volume sampling removes the logarithmic factor and selects exactly the information-theoretic minimum $\binom{n}{s}$ columns with probability one; for fixed s in a middle layer this is exponentially sparse. The same minimal projection-DPP construction applies to bounded-alphabet product degree spaces, whose dimension is $O_{Q,s}(n^s)$ against an

ambient domain of size at least 2^n . At degree one, twice-degree unisolvence removes any ambient-extension assumption: every fixed degree-two-unisolvent Johnson column family simultaneously satisfies the adjacent-two-value theorem for every row family. Such a family exists with exactly $\binom{n}{2}$ columns, for example in the explicit radius-two cross of Theorem 4.18; projection-DPP sampling gives an invariant alternative. It satisfies

$$D(M) \leq \lceil \log_2(\text{rank } M + 1) \rceil + 1, \quad \text{rect}(M) \geq 1/\lceil 4(\text{rank } M + 1) \rceil.$$

On every fixed balanced range $\beta n \leq k \leq (1 - \beta)n$, a quadratic-scale hypergeometric two-point escape theorem reduces the universal column count further to $O_\beta(n)$, with the same bounds. A strengthened construction needs no protected rank anchor: the same random column family is simultaneously a two-value detector and a robust degree-one interpolation set. It retains the theorem after adversarial deletion of a $\beta^2/10$ -fraction of *all* its columns. The underlying hypergeometric escape constant is exactly $2/5$ after quadratic normalization, with unique equality at the central $J(6, 3)$ three-marked distribution. There is also a deterministic spectral version: any k -uniform column family whose empirical incidence covariance has eigenvalue greater than $n/[8(n - 2)]$ on $\mathbf{1}^\perp$ satisfies the same logarithmic protocol. Balanced 2-designs meet this condition in a broad parameter range; the Sylvester–Hadamard construction gives an explicit family of exactly n columns for every $n = 2^t - 1$, $t \geq 3$.

As a concrete application, partition $[d]$ into arbitrary blocks and let the right family choose a fixed number k_i from each block, where $2 \leq k_i \leq n_i - 2$. This deterministic family can have exponentially small relative density in the global layer and is generally not a complete affine-code section. Nevertheless every adjacent two-value opposite family has the exact row classification: every nonconstant row is a coordinate function or its complement. This gives a monochromatic rectangle of relative density at least $1/\lceil 4(d + 1) \rceil$ and the rank-sensitive communication bound $D(M) \leq \lceil \log_2(\text{rank } M + 1) \rceil + 4$, which is optimal up to an additive constant. We also prove a robust transfer from a nearby rational subspace, with explicit projector, Boolean-margin, and integer-height losses. For layers at the boundary $k_i = 1$ or $n_i - 1$, arbitrary Boolean matrices can be embedded, so the interior hypothesis marks a sharp barrier. More general blockwise Johnson layers admit a bound governed by an explicit weighted Boolean tube radius. We also prove a logarithmic rank-sensitive protocol for products of interior multislices over a bounded alphabet, with the same sharp count-one boundary. These are restricted-column results and do not solve the general Log-Rank Conjecture.

1 Introduction

The Log-Rank Conjecture asks whether the deterministic communication complexity of a Boolean matrix is polynomially bounded by the logarithm of its real rank. Equivalently, it predicts a quasipolynomial relation between real rank and the partition number of a Boolean matrix. The strongest known general bound gives a monochromatic rectangle of relative density $2^{-O(\sqrt{r})}$ in every rank- r Boolean matrix [ST25, HLS25].

Hambardzumyan, Lovett, and Shirley recently showed that Log-Rank is equivalent to a relative rectangle conjecture for two set families whose cross-intersections take two adjacent values [HLS25]. In that formulation one has families $\mathcal{A}, \mathcal{B} \subseteq 2^{[d]}$ and an integer a such that

$$|A \cap B| \in \{a, a + 1\} \quad (A \in \mathcal{A}, B \in \mathcal{B}). \quad (1)$$

The Boolean matrix

$$M_{A,B} = \mathbf{1}_{\{|A \cap B| = a+1\}} \quad (2)$$

has real rank at most $d + 1$. The conjecture asks for a monochromatic rectangle retaining a $2^{-\text{polylog } d}$ fraction of all cross-pairs.

The problem remains difficult precisely when both families are sparse. Classical and modern anti-concentration theorems control dense cube families or independent coordinates [Erd45,

RY22, TV09, NV11]; the binary-scalar-product theorems of Kupavskii–Weltge and Kupavskii–Tsarev control absolute products of full-span configurations [KW22, KT26]. Neither type of conclusion is a relative rectangle theorem for an arbitrarily sparse prescribed column family.

Filmus–Ihringer classify Boolean degree-one functions on an individual interior Johnson scheme as constants, dictators, and their complements [FI19b]. Our local interval argument recovers this set-case fact. The contributions here are the product assembly, the cross-intersection interpretation, the rank-sensitive protocol and sharp boundary universality, and the spectral–lattice extension.

Our first contribution is a general sufficient condition formulated through the covariance of the right family and the arithmetic geometry of a low-mode subspace. It is useful even when that subspace has dimension linear in d . Our second contribution is a particularly clean natural application. A third contribution is a general product transfer principle that converts factorwise degree-one classifications and atom-rank growth into Log-Rank protocols. A fourth is a rank-pooling principle for low-degree Boolean rows: a row-space basis compresses the common coordinate universe, after which low-depth query trees become rank-sensitive communication protocols. A fifth is an anchor-free sparse transfer: one linear-size sample simultaneously detects every forbidden two-value row and remains degree-one unisolvent after adversarial column deletion. A sixth replaces randomness entirely by a deterministic empirical covariance gap and includes explicit linear-size Hadamard-design columns.

Theorem 1.1 (Interior product-Johnson theorem; informal). *Partition $[d]$ into blocks of sizes n_i , and let $\mathcal{Y}$ consist of all sets containing exactly k_i points from block i , where $2 \leq k_i \leq n_i - 2$. If an arbitrary family $\mathcal{A}$ satisfies (1) with $\mathcal{Y}$, then the matrix (2) satisfies*

$$\text{rank}_{\mathbb{R}} M \leq d + 1, \quad \text{rect}(M) \geq \frac{1}{4(d+1)}, \quad D(M) \leq \lceil \log_2(\text{rank } M + 1) \rceil + 4.$$

Theorem 1.2 (Anchor-free sparse HLS; informal). *Fix $0 < \beta \leq 1/2$. On every layer $\beta n \leq k \leq (1 - \beta)n$, a uniform random family of $O_{\beta}(n + \log(1/\delta))$ columns has, with probability at least $1 - \delta$, the following simultaneous property. After arbitrary deletion of at most a $\beta^2/10$ -fraction of all sampled columns, every adjacent-two-value set-row matrix on the retained family satisfies*

$$D(M) \leq \lceil \log_2(\text{rank}_{\mathbb{R}} M + 1) \rceil + 1, \quad \text{rect}(M) \geq \frac{1}{4(\text{rank}_{\mathbb{R}} M + 1)}.$$

No subset of columns is designated as a protected rank anchor.

Theorem 1.3 (Deterministic spectral sparse HLS; informal). *Let $Y \subseteq \binom{[n]}{k}$ be nonempty and let Σ_Y be the empirical covariance matrix of its incidence vectors. If, for every $v \perp \mathbf{1}$,*

$$v^{\top} \Sigma_Y v \geq \lambda \|v\|_2^2, \quad \lambda > \frac{n}{8(n-2)},$$

then every adjacent-two-value set-row matrix on Y satisfies

$$D(M) \leq \lceil \log_2(\text{rank}_{\mathbb{R}} M + 1) \rceil + 1, \quad \text{rect}(M) \geq \frac{1}{4(\text{rank}_{\mathbb{R}} M + 1)}.$$

This is a deterministic condition on the prescribed column family. In particular, explicit Sylvester–Hadamard designs give $|Y| = n$ on the infinite sequence $n = 2^t - 1$.

Theorem 1.4 (Low-degree slice theorem; informal). *Let M be a Boolean matrix whose columns are indexed by $\binom{[n]}{k}$, and suppose every row has slice degree at most $s \geq 1$. If $\min(k, n-k) \geq 144s^4$ and $r = \text{rank}_{\mathbb{R}} M$, then, with $q = 5184s^8$,*

$$D(M) \leq q(\lceil \log_2(r+1) \rceil + q + 1).$$

In particular, the Log-Rank Conjecture holds for this class whenever $s = \text{polylog}(r+1)$. For each fixed s , the qualitative conclusion $D(M) = O_s(\log(r+1))$ already holds under the sharp junta condition $\min(k, n-k) \geq 2s$.

The interior condition is sharp in a complexity-theoretic sense. If a block with $k_i = 1$ is allowed, its columns are singletons and arbitrary Boolean matrices can be encoded by taking each row to be its support. Thus extending the logarithmic theorem to the boundary would include the general Log-Rank Conjecture.

Using the Filmus–Ihringer classification on multislices, the same product and rank method gives a logarithmic protocol for products of multislices in which every color has multiplicity at least two. A color of multiplicity one again embeds arbitrary Boolean matrices.

For fixed block parameters and a growing number of blocks, $\mathcal{Y}$ has exponentially small relative density in the global middle layer. It is deterministic and is generally not the intersection of that layer with a binary affine coset. Thus Theorem 1.1 is not a dense-subset, random-thinning, or complete-affine-section corollary.

The proof has two complementary descriptions. An elementary blockwise hypergeometric variance computation shows that every row is within one coordinate of a union of whole blocks. More conceptually, the covariance vanishes on the span of block indicators and has a constant gap on its orthogonal complement; the Boolean tube at the two-point variance threshold has Hamming radius one.

Our general theorem replaces the block span by an arbitrary integer-generated real subspace. The main parameters are an exact Boolean tube radius and the entropy of the integer generator statistics. A perturbative version permits the true low-mode space to be merely close to the rational one.

Claim boundary. All results in this paper are restricted theorems. They do not improve the best known bound for every rank- r Boolean matrix. The product-Johnson class does admit a full logarithmic protocol; the general spectral-lattice theorem is only a one-rectangle result. The publication-level novelty comparison is still subject to independent expert verification.

2 Two-level matrices and rectangle density

For nonempty finite sets X, Y and $M \in \{0, 1\}^{X \times Y}$, define

$$\text{rect}(M) = \max_{X_0 \times Y_0 \text{ monochromatic}} \frac{|X_0|}{|X|} \frac{|Y_0|}{|Y|}.$$

Write $D(M)$ for deterministic communication complexity. We use the one-party-output convention; requiring both parties to output changes all bounds below by at most one bit.

Under (1),

$$M_{A,B} = |A \cap B| - a = -a + \sum_{j=1}^d \mathbf{1}_{\{j \in A\}} \mathbf{1}_{\{j \in B\}}. \quad (3)$$

Hence

$$\text{rank}_{\mathbb{R}} M \leq d + 1. \quad (4)$$

We repeatedly use the following elementary fact.

Lemma 2.1 (Two-point variance). *If a real random variable is supported on two consecutive integers, then its variance is at most $1/4$.*

Proof. After translation it is Bernoulli with parameter p , and its variance is $p(1-p) \leq 1/4$. $\square$

3 An atomic product transfer theorem

We isolate the communication argument from the particular combinatorial domains used later. Let Ω_i be finite sets and let $L_i \subseteq \mathbb{R}^{\Omega_i}$ be real linear spaces containing the constant function. On $\Omega = \prod_i \Omega_i$, define the additive degree-one space

$$L = \left\{ c + \sum_i f_i(x_i) : c \in \mathbb{R}, f_i \in L_i \right\}.$$

For each factor, let $\mathcal{Q}_i$ be a set of nonconstant Boolean atoms, with one representative from each complementary pair. For a nonempty finite $\mathcal{F} \subseteq \mathcal{Q}_i$, define its centered rank

$$\rho_i(\mathcal{F}) = \dim \text{span}(\{1\} \cup \mathcal{F}) - 1.$$

Theorem 3.1 (Atomic product transfer). *Suppose the following hold for fixed $C \geq 1$ and $\alpha \geq 1$:*

1. *every nonconstant Boolean function in L_i is an atom in $\mathcal{Q}_i$ or its complement;*
2. *every nonempty finite $\mathcal{F} \subseteq \mathcal{Q}_i$ satisfies $|\mathcal{F}| \leq C \rho_i(\mathcal{F})^\alpha$.*

Then every Boolean matrix M with column set Ω and rows in L satisfies

$$D(M) \leq \lceil \log_2(2C(\text{rank}_{\mathbb{R}} M + 1)^\alpha + 2) \rceil.$$

In particular, $D(M) \leq \alpha \log_2(\text{rank } M + 1) + O(\log C + 1)$.

Proof. Write a Boolean row as $f = c + \sum_i f_i(x_i)$. Fixing all factors except i shows that the range of every nonconstant f_i has exactly two values. If two factors were nonconstant, the sum of their two two-point ranges would contain at least three values, contradicting Booleanity. Thus every nonconstant row is a lifted atom from one factor or its complement.

Let $\mathcal{F}_i$ be the positive atoms used by the matrix rows and put $K = \sum_i |\mathcal{F}_i|$. Replacing complements by positive atoms enlarges the row span by at most the constant function. Hence all lifted atoms lie in a space whose quotient by constants has dimension at most $r = \text{rank}_{\mathbb{R}} M$.

The centered one-factor spaces form a direct sum on the product: if $\sum_i g_i(x_i)$ is constant, varying one coordinate shows that every g_i is constant. Consequently

$$\sum_i \rho_i(\mathcal{F}_i) \leq r.$$

The atom-growth hypothesis and convexity give

$$K \leq C \sum_i \rho_i(\mathcal{F}_i)^\alpha \leq Cr^\alpha.$$

Alice sends one of two constant messages, or the used atom index and a complement bit. There are at most $2K + 2$ messages, proving the theorem. $\square$

The polynomial atom-growth hypothesis has a natural opposite obstruction.

Proposition 3.2 (Selector universality). *Suppose one factor contains Boolean degree-one functions $q_1, \dots, q_N$ satisfying*

$$q_i q_j = 0 \ (i \neq j), \quad \sum_{j=1}^N q_j = 1,$$

and every selector value j occurs on the domain. Then row-degree-one Boolean matrices on that factor contain every Boolean matrix with N column types.

Proof. Choose one domain point x_j with $q_j(x_j) = 1$. For an arbitrary Boolean row vector $v \in \{0, 1\}^N$, the degree-one function

$$f_v = \sum_{j: v_j=1} q_j$$

is Boolean and satisfies $f_v(x_j) = v_j$. Applying this construction to every row embeds the prescribed matrix. Extra domain points only duplicate one of the selector column types. $\square$

Thus polynomial centered-rank atom growth is a sufficient tractable regime, while a large selector partition recovers unrestricted Boolean matrices.

4 Low-degree cube and slice matrices

We next give a second transfer mechanism which is not limited to degree one. For a Boolean function on the cube, degree means the degree of its unique multilinear real polynomial. For a function on $\binom{[n]}{k}$, slice degree is the minimum degree of a real polynomial agreeing with it on the layer.

We use three published ingredients. Filmus–Ihringer prove that there is a universal $C_0 > 1$ such that, when $\min(k, n - k) \geq C_0^s$, every Boolean slice function of degree at most s is the restriction of a Boolean cube function of the same degree [FI19a]. Chiarelli–Hatami–Saks prove that every Boolean cube function of degree at most s depends on at most $6.614 \cdot 2^s$ coordinates [CHS20]. Finally, Midrijanis proves that such a function has deterministic decision-tree depth at most $2s^3$ [Mid04]. Put

$$J_s = 7 \cdot 2^s, \quad q_s = 2s^3,$$

and enlarge C_0 , if necessary, so that $C_0^s \geq J_s$ for every $s \geq 1$.

The new point is that the coordinate names used by the query trees can be compressed in terms of the actual matrix rank.

Lemma 4.1 (Rank pooling on the cube). *Let M be a Boolean matrix of rank $r \geq 1$ whose columns are indexed by $\{0, 1\}^n$. If every row depends on at most J coordinates, then all rows depend on one common set of at most Jr coordinates.*

Proof. Choose r rows $f_1, \dots, f_r$ forming a basis of the row space, choose a relevant-coordinate set S_i of size at most J for each one, and put $S = \bigcup_i S_i$. If two columns agree on S , every basis row takes the same value on them. Every other row is a real linear combination of the basis rows and therefore also takes the same value. Thus every row depends on S , while $|S| \leq Jr$. $\square$

Theorem 4.2 (Low-degree cube Log-Rank theorem). *Let M be a Boolean matrix whose columns are indexed by $\{0, 1\}^n$, and suppose every row has degree at most $s \geq 1$. If $r = \text{rank}_{\mathbb{R}} M$, then*

$$D(M) \leq 2s^3 (\lceil \log_2(7 \cdot 2^s(r + 1)) \rceil + 1).$$

Proof. If $r = 0$, the matrix is identically zero. Otherwise apply Lemma 4.1 with $J = J_s$. For each row, Alice simulates a depth- q_s decision tree. At every internal node she sends the index, inside the common coordinate pool, of the next queried coordinate; Bob returns its bit. Each round costs at most $\lceil \log_2(J_s r) \rceil + 1$ bits. Alice knows the reached leaf and outputs its label. Replacing r by $r + 1$ gives the displayed uniform bound. $\square$

The only extra issue on a slice is that junta supports need not be unique, because the total weight is fixed. The following synchronization lemma handles exactly this point.

Lemma 4.3 (Slice support intersection). *Let $f : \binom{[n]}{k} \rightarrow \mathbb{R}$ be both an S -junta and a T -junta. If*

$$[n] \setminus (S \cup T) \neq \emptyset,$$

then f is an $(S \cap T)$ -junta.

Proof. Put $U = S \cap T$ and choose $j \notin S \cup T$. We first show that f is invariant under every transposition of two coordinates outside U . A transposition of two points outside S preserves the S -trace and one of two points outside T preserves the T -trace. The only remaining case, after exchanging the two names, has $a \in S \setminus T$ and $b \in T \setminus S$. The identity

$$(a \ b) = (a \ j)(b \ j)(a \ j)$$

expresses it as two transpositions outside T and one outside S , so it also preserves f .

If two k -sets have the same trace on U , their two symmetric-difference halves have the same size and lie outside U . Pair the two halves and swap the paired coordinates. The resulting sequence of transpositions outside U carries one set to the other, and every step preserves f . Hence f depends only on its U -trace. $\square$

This lemma combines with the sharp fixed-degree junta threshold on the slice.

Theorem 4.4 (Optimal-threshold fixed-degree slice theorem). *For every $s \geq 1$ there is an integer $m_s \geq 1$ such that the following holds. If*

$$\min(k, n - k) \geq 2s$$

and every row of a Boolean matrix M on $\binom{[n]}{k}$ has slice degree at most s , then, writing $r = \text{rank}_{\mathbb{R}} M$,

$$D(M) \leq \lceil \log_2 ((m_s + 1)(m_s(r + 1) + 1)^{m_s} 2^{2^{m_s}}) \rceil.$$

In particular, $D(M) = O_s(\log(r + 1))$.

Proof. Filmus proves that every Boolean degree- s function on such a slice is an m_s -junta, for a constant m_s depending only on s [Fil22]. His theorem is stated for $k \leq n/2$; the other half follows by complementing every coordinate, which preserves degree.

Choose r basis rows and let S be the union of junta supports of size at most m_s for those rows. Then $|S| \leq m_s r$, and every matrix row is an S -junta. An arbitrary row also has some junta support T with $|T| \leq m_s$. If $n > m_s(r + 1)$, then $S \cup T \neq [n]$, and Lemma 4.3 replaces T by $S \cap T$. If $n \leq m_s(r + 1)$, use all of $[n]$. Thus in both cases every row has a junta support of size at most m_s inside one common universe U with

$$|U| \leq m_s(r + 1).$$

There are at most

$$\sum_{j=0}^{m_s} \binom{|U|}{j} 2^{2^j} \leq (m_s + 1)(|U| + 1)^{m_s} 2^{2^{m_s}}$$

distinct rows: choose a support and then a Boolean truth table on it. Alice sends the index of her row among these possibilities, and Bob evaluates it on his column. The displayed count proves the theorem. $\square$

The preceding theorem uses a non-explicit junta-size constant. A polynomial layer-depth assumption permits a completely explicit bound. The extra ingredient is a symmetry lemma for decision trees.

Lemma 4.5 (Symmetric-tree canonicalization). *Let $f : \binom{[n]}{k} \rightarrow \{0, 1\}$ be an S -junta and suppose f has a coordinate decision tree of depth at most q . There is a set $R \subseteq [n] \setminus S$ with $|R| \leq q$ and a depth- q decision tree for f all of whose queried coordinates lie in $S \cup R$.*

Proof. Choose any $R \subseteq [n] \setminus S$ of size $\min\{q, |[n] \setminus S|\}$. We prove by induction on q that every depth- q tree can be replaced by one using only $S \cup R$.

There is nothing to prove for a leaf. Otherwise let i be the root query. If $i \notin S$ and $i \notin R$, choose $j \in R$ and conjugate the entire tree by the transposition $(i\ j)$. This still computes f , since the transposition fixes S and f is an S -junta, and its root query is now j . Thus the root lies in $S \cup R$.

If the root is a point of R , remove it from R for the recursive calls and add it to the fixed support; each restricted function is invariant under all permutations outside S together with that root. If the root lies in S , use the same fixed subset of at most $q - 1$ points of R in both recursive calls. Conjugating subtrees in the same way proves the induction step. $\square$

We can now state the general communication principle. A coordinate domain $\Omega \subseteq [Q]^n$ is called symmetric if it is invariant under every permutation of its n coordinates.

Theorem 4.6 (Rank-pooled bounded-answer query transfer). *Let M be a Boolean matrix of real rank r whose column domain is either*

1. *a symmetric subdomain of $[Q]^n$, or*
2. *a complete product $\prod_{i=1}^n \Sigma_i$ with $2 \leq |\Sigma_i| \leq Q$.*

Suppose every row function has a coordinate decision tree of depth at most q . Put $L = \lceil \log_2 Q \rceil$. Then

$$D(M) \leq q(\lceil \log_2(r+1) \rceil + (q+1)L).$$

Consequently, if both q and $\log Q$ are polynomial in $\log(r+1)$, then the Log-Rank Conjecture holds for M .

Proof. Assume $r \geq 1$ and choose r basis rows together with depth- q coordinate trees. A Q -ary depth- q tree has fewer than Q^q internal nodes. If S is the union of every coordinate queried by the basis trees, then

$$|S| \leq rQ^q.$$

Every matrix row is a real linear combination of the basis rows and hence is an S -junta.

In the symmetric case, Lemma 4.5, whose proof is unchanged for a multiway tree, puts every row tree inside one common universe $U = S \cup R$ with $|R| \leq q$. Thus

$$|U| \leq rQ^q + q \leq (r+1)Q^q.$$

In the complete-product case, every query outside S can instead be pruned: all its answer restrictions compute the same residual function, and any one child is valid on every assignment of the remaining coordinates. Hence one may take $U = S$.

Alice simulates the resulting row tree. Each round she names a coordinate of U , using at most $\lceil \log_2(r+1) \rceil + qL$ bits, and Bob returns its value using L bits. There are at most q rounds. $\square$

Dafni, Filmus, Lifshitz, Lindzey, and Vinyals prove quantitative polynomial relations between degree and coordinate decision-tree depth on multislices [DFL⁺21]. Their parameters give the following explicit slice consequence. If

$$\min(k, n-k) \geq 144s^4,$$

then every Boolean slice function of degree at most s has decision-tree depth at most

$$q_s = 5184s^8. \tag{5}$$

Indeed, in their notation a slice has $\Delta = 1$, conflict bound $M = \lfloor \min(k, n-k)/2 \rfloor$, and block-sensitivity ratio $\beta \geq 1/2$. Their Theorem 3.1 gives $\text{bs}(f) \leq 6s^2$ and, once $s \leq (\beta M/36)^{1/4}$,

$$D_{\text{query}}(f) \leq \beta^{-2} \Delta \text{bs}(f)^4 \leq 4(6s^2)^4 = 5184s^8.$$

Theorem 4.7 (Polynomial-depth low-degree slice Log-Rank theorem). *Let M be a Boolean matrix whose columns are indexed by $\binom{[n]}{k}$ and whose rows have slice degree at most $s \geq 1$. If*

$$\min(k, n - k) \geq 144s^4,$$

put $r = \text{rank}_{\mathbb{R}} M$ and $q_s = 5184s^8$. Then

$$D(M) \leq q_s (\lceil \log_2(r + 1) \rceil + q_s + 1).$$

In particular,

$$D(M) = O(s^8 \log(r + 1) + s^{16}).$$

Proof. The Johnson slice is a symmetric binary coordinate domain. Equation (5) supplies depth q_s for every row. Apply Theorem 4.6 with $Q = 2$ and $L = 1$. $\square$

Theorem 4.8 (Very-deep slice refinement). *There is a universal constant $C > 1$ such that the following holds. Let M be a Boolean matrix whose columns are indexed by $\binom{[n]}{k}$, and suppose every row has slice degree at most $s \geq 1$. If*

$$\min(k, n - k) \geq C^s$$

and $r = \text{rank}_{\mathbb{R}} M$, then

$$D(M) \leq 2s^3 (\lceil \log_2(7 \cdot 2^s(r + 1)) \rceil + 1).$$

Consequently

$$\text{rect}(M) \geq 2^{-2s^3(\lceil \log_2(7 \cdot 2^s(r+1)) \rceil + 1)}.$$

Proof. The rank-zero case is immediate, so assume $r \geq 1$. Choose a row-space basis $f_1, \dots, f_r$. Extend each f_i to a Boolean degree- s cube function and let S_i be a relevant-coordinate set of size at most J_s . Put $S = \bigcup_i S_i$, so $|S| \leq J_s r$. On the slice, every matrix row is a linear combination of the f_i and hence is an S -junta.

For an arbitrary row f , choose its own Boolean degree- s cube extension g and a relevant set T with $|T| \leq J_s$. If $n > J_s(r + 1)$, then $[n] \setminus (S \cup T)$ is nonempty. Lemma 4.3 shows that the restriction of g to the slice is an $(S \cap T)$ -junta. Every Boolean pattern on T occurs on the slice because $|T| \leq J_s \leq \min(k, n - k)$. It follows that the total cube function g itself is independent of $T \setminus S$. Thus its decision tree queries only coordinates in S .

If instead $n \leq J_s(r + 1)$, use all of $[n]$ as the coordinate pool. In both cases every row has a depth- q_s decision tree whose queried coordinates lie in one common universe U satisfying

$$|U| \leq J_s(r + 1).$$

The protocol from Theorem 4.2 now proves the communication bound. A deterministic c -bit protocol has at most 2^c transcript rectangles, so one monochromatic rectangle has relative density at least 2^{-c} . $\square$

Theorem 4.4 gives $D(M) = O_s(\log(r + 1))$ already at the optimal junta threshold. The polynomial-depth Theorem 4.7 gives a fully explicit polylogarithmic bound whenever $s = \text{polylog}(r + 1)$, under the polynomial layer-depth condition $144s^4$. In the still deeper C^s regime, Theorem 4.8 improves the dependence on s from degrees eight and sixteen to degrees three and four. All layer-depth conditions concern only the ambient slice and are independent of matrix rank.

The opposite boundary is universal.

Proposition 4.9 (Low-degree boundary universality). *If $k \leq s$, every Boolean function on $\binom{[n]}{k}$ has slice degree at most s . The same holds if $n - k \leq s$. In particular, the case $k = 1$ contains every Boolean matrix with n columns.*

Proof. For $\mathcal{F} \subseteq \binom{[n]}{k}$, its indicator on the layer is

$$\mathbf{1}_{\mathcal{F}}(x) = \sum_{A \in \mathcal{F}} \prod_{i \in A} x_i.$$

Exactly one degree- k monomial is one at each layer point. This proves the first assertion; the second follows by replacing every coordinate by its complement. When $k = 1$, the columns are the coordinate unit vectors and an arbitrary row is the indicator of an arbitrary subset of them. $\square$

The threshold $2s$ in Theorem 4.4 is sharp for a uniform bounded-junta conclusion [Fil22]. The polynomial $144s^4$ hypothesis supplies an explicit shallow decision tree, while the larger C^s hypothesis supplies a same-degree total-cube extension and the sharper dependence on s .

4.1 Bounded alphabets and multislices

The rank-pooling mechanism extends beyond binary slices. Let

$$\Omega_{\text{prod}} = \prod_{i=1}^n [m_i], \quad 2 \leq m_i \leq Q,$$

and define degree using the coordinate-value indicators $\mathbf{1}_{\{x_i=a\}}$. A query asks for x_i and has at most Q answers.

Theorem 4.10 (Low-degree bounded-alphabet product theorem). *Let M be a Boolean matrix with column domain Ω_{prod} and rows of degree at most $s \geq 1$. Put*

$$r = \text{rank}_{\mathbb{R}} M, \quad q = 1296s^8, \quad L = \lceil \log_2 Q \rceil.$$

Then

$$D(M) \leq q (\lceil \log_2(r+1) \rceil + (q+1)L).$$

For fixed Q , this is $O_Q(s^8 \log(r+1) + s^{16})$.

Proof. For a complete product domain, the parameters in [DFL⁺21, Theorem 3.1] are $\Delta = 1$, conflict bound $M_0 = n$, and $\beta = 1$. Hence every Boolean degree- s row has coordinate decision-tree depth at most

$$\text{bs}(f)^4 \leq (6s^2)^4 = 1296s^8 = q.$$

Apply Theorem 4.6 in its complete-product case. $\square$

The bounded-alphabet condition is essential. A single coordinate with a large alphabet is a selector domain: its degree-one Boolean functions already encode arbitrary Boolean matrices.

Now let $\lambda = (\lambda_1, \dots, \lambda_m)$ be positive integers summing to n , with $m \leq Q$, and consider the multislice

$$\Omega(\lambda) = \{x \in [m]^n : |\{i : x_i = c\}| = \lambda_c \text{ for every } c\}.$$

Write $\lambda_{\min} = \min_c \lambda_c$.

Theorem 4.11 (Low-degree multislice Log-Rank theorem). *Let M be a Boolean matrix with column domain $\Omega(\lambda)$ and rows of multislice degree at most $s \geq 1$. Assume*

$$\lambda_{\min} \geq 216s^4.$$

Put

$$r = \text{rank}_{\mathbb{R}} M, \quad q = 11664s^8, \quad L = \lceil \log_2 Q \rceil.$$

Then

$$D(M) \leq q (\lceil \log_2(r+1) \rceil + (q+1)L).$$

For fixed Q , this is $O_Q(s^8 \log(r+1) + s^{16})$.

Proof. For a multislice, the parameters in [DFL⁺21, Theorem 3.1 and Section 4.3] are

$$\Delta = 1, \quad M_0 = \lfloor \lambda_{\min}/2 \rfloor, \quad \beta \geq 1/3.$$

The layer hypothesis implies $s \leq (\beta M_0/36)^{1/4}$. Their strengthened bound therefore gives

$$D_{\text{query}}(f) \leq \beta^{-2} \text{bs}(f)^4 \leq 9(6s^2)^4 = 11664s^8 = q.$$

The multislice is symmetric under coordinate permutations. Apply Theorem 4.6. $\square$

When a color has multiplicity one, degree-one selector universality already recovers arbitrary Boolean matrices. Thus some lower bound on $\lambda_{\min}$ is structurally unavoidable, although the polynomial $216s^4$ threshold is not claimed to be sharp.

4.2 Unisolvent sparse-column transfer

The preceding theorems remain valid after a sparse column restriction that preserves the ambient function space. Let $V \subseteq \mathbb{R}^\Omega$ be a finite-dimensional space and $Y \subseteq \Omega$. Call Y *V-unisolvent* if the restriction map

$$\text{res}_Y : V \longrightarrow \mathbb{R}^Y$$

is injective.

Theorem 4.12 (Unisolvent sparse-column transfer). *Assume Ω satisfies one of the two domain hypotheses in Theorem 4.6, every Boolean function in V has coordinate-query depth at most q , and Y is V -unisolvent. Then every Boolean matrix M with columns Y whose rows extend to Boolean functions in V satisfies*

$$D(M) \leq q(\lceil \log_2(\text{rank}_{\mathbb{R}} M + 1) \rceil + (q+1)\lceil \log_2 Q \rceil).$$

Proof. Let $\widetilde{M}$ be the matrix with the same row functions and all ambient columns Ω . A row relation of $\widetilde{M}$ remains one after restriction. Conversely, a restricted relation is an element of V vanishing on Y , and unisolvence forces it to vanish on Ω . Hence

$$\text{rank}_{\mathbb{R}} \widetilde{M} = \text{rank}_{\mathbb{R}} M.$$

Apply Theorem 4.6 to $\widetilde{M}$ and restrict Bob's inputs to Y . $\square$

The assumption that each sparse row already has a Boolean ambient extension can itself be certified by unisolvence at twice the degree.

Theorem 4.13 (Booleanity-certifying sparse transfer). *Let $V_{\leq s} \subseteq V_{\leq 2s} \subseteq \mathbb{R}^\Omega$ be a degree filtration such that*

$$f \in V_{\leq s} \implies f(f-1) \in V_{\leq 2s}.$$

If Y is $V_{\leq 2s}$ -unisolvent and $f \in V_{\leq s}$ takes values in $\{0, 1\}$ on Y , then f is Boolean on all of Ω . Consequently Theorem 4.12 applies to Boolean matrices on Y whose rows merely extend to real functions in $V_{\leq s}$.

Proof. The function $h = f(f-1)$ belongs to $V_{\leq 2s}$ and vanishes on Y . Unisolvence gives $h = 0$ on Ω , so $f(x) \in \{0, 1\}$ everywhere. Moreover $V_{\leq s} \subseteq V_{\leq 2s}$, so every restricted row relation lifts to the ambient domain as in Theorem 4.12. $\square$

We next show that random sparse subsets are unisolvent on every transitive invariant function space. Let a finite group act transitively on Ω , let $V \subseteq \mathbb{R}^\Omega$ be invariant, and put $d = \dim V$. Choose an orthonormal basis $\phi_1, \dots, \phi_d$ for the uniform inner product and write

$$z_x = (\phi_1(x), \dots, \phi_d(x))^\top.$$

The diagonal of the reproducing kernel is invariant and hence constant. Taking its average gives

$$\|z_x\|_2^2 = d, \quad \mathbb{E}_{x \in \Omega} z_x z_x^\top = I_d. \quad (6)$$

Theorem 4.14 (Random unisolvent sampling). *Let Y be a uniformly random N -element subset of Ω , sampled without replacement. Then*

$$\Pr[Y \text{ is not } V\text{-unisolvent}] \leq d \exp\left(-\frac{N}{8d}\right).$$

Consequently $N \geq 8d \log(d/\delta)$ gives unisolvence with probability at least $1 - \delta$.

Proof. Restriction is injective exactly when

$$G_Y = \sum_{x \in Y} z_x z_x^\top$$

is positive definite. By (6), the summands are positive semidefinite, bounded by dI_d , and have mean I_d . Matrix Chernoff at lower deviation $1/2$ [Tro12], together with the without-replacement Laplace-transform comparison of Gross–Nesme [GN10], yields

$$\Pr[\lambda_{\min}(G_Y) \leq N/2] \leq d \left(\sqrt{\frac{2}{e}}\right)^{N/d} \leq d e^{-N/(8d)}.$$

Outside this event G_Y is positive definite. □

The logarithmic oversampling factor can be removed by standard projection-DPP volume sampling. Let $A \in \mathbb{R}^{|\Omega| \times d}$ have orthonormal columns spanning the evaluation image of V in the standard Euclidean inner product, and put $P = AA^\top$.

Theorem 4.15 (Minimal projection-DPP unisolvent sampling). *For every d -element subset $Y \subseteq \Omega$, define*

$$\Pr_{\text{DPP}}[Y] = \det(P_Y).$$

These probabilities sum to one, and every set in their support is V -unisolvent. Thus the process always selects exactly the information-theoretic minimum $d = \dim V$ columns.

Proof. For $|Y| = d$,

$$\det(P_Y) = \det(A_Y A_Y^\top) = \det(A_Y)^2.$$

Cauchy–Binet gives

$$\sum_{|Y|=d} \det(P_Y) = \sum_{|Y|=d} \det(A_Y)^2 = \det(A^\top A) = 1.$$

Positive probability is equivalent to A_Y being nonsingular, which is exactly unisolvence. Conversely, injectivity of $V \rightarrow \mathbb{R}^Y$ requires $|Y| \geq d$. □

When V is invariant under a transitive group, P and the resulting projection DPP are invariant. Its one-point marginals are uniform:

$$\Pr[x \in Y] = P_{xx} = d/|\Omega|.$$

For the Johnson slice, let $V_{\leq s}$ be the space of functions of slice degree at most s . The harmonic decomposition gives

$$\dim V_{\leq s} = \sum_{j=0}^s \left(\binom{n}{j} - \binom{n}{j-1} \right) = \binom{n}{s} \quad (s \leq \min(k, n-k)) \quad (7)$$

[Fil16], with $\binom{n}{-1} = 0$. This space is invariant under the transitive S_n -action.

Corollary 4.16 (Exponentially sparse random Johnson columns). *Assume*

$$\min(k, n - k) \geq 144s^4, \quad d_s = \binom{n}{s},$$

and choose a uniformly random N -element family $Y \subseteq \binom{[n]}{k}$ with

$$N \geq 8d_s \log(d_s/\delta).$$

With probability at least $1 - \delta$, simultaneously for every Boolean matrix whose rows extend to degree- s Boolean functions on the full slice,

$$D(M) \leq q_s (\lceil \log_2(\text{rank}_{\mathbb{R}} M + 1) \rceil + q_s + 1), \quad q_s = 5184s^8.$$

Proof. Theorem 4.14 and (7) make Y $V_{\leq s}$ -unisolvent with the stated probability. Equation (5) supplies query depth q_s , so Theorem 4.12 applies with $Q = 2$. The unisolvence event depends only on Y , which gives the simultaneous statement over all later row families. $\square$

Corollary 4.17 (Minimal projection-DPP Johnson columns). *Under the hypotheses of Corollary 4.16, sample Y from the rank- $\binom{n}{s}$ projection DPP of $V_{\leq s}$. Then*

$$|Y| = \binom{n}{s}$$

and, with probability one, the communication conclusion of that corollary holds simultaneously for every Boolean degree- s row family restricted to Y . No smaller column set can be unisolvent for the entire space $V_{\leq s}$.

Proof. Apply Theorem 4.15, then Theorem 4.12. $\square$

The twice-degree certificate gives a particularly sharp adjacent-two-value theorem with only quadratically many columns.

Theorem 4.18 (Explicit quadratic unisolvent cross). *Assume $2 \leq k \leq n - 2$. Fix a k -set I , put $O = [n] \setminus I$, and choose fixed pairs $I_0 \in \binom{I}{2}$ and $O_0 \in \binom{O}{2}$. Define*

$$\begin{aligned} \mathcal{Y}_{\times} = & \{I\} \cup \{I \setminus \{i\} \cup \{a\} : i \in I, a \in O\} \\ & \cup \{I \setminus \{i, j\} \cup O_0 : \{i, j\} \in \binom{I}{2}\} \\ & \cup \{I \setminus I_0 \cup \{a, b\} : \{a, b\} \in \binom{O}{2}\}. \end{aligned}$$

Then

$$|\mathcal{Y}_{\times}| = \binom{n}{2},$$

and $\mathcal{Y}_{\times}$ is unisolvent for the degree-at-most-two slice space.

Proof. The two double-swap families overlap only at $I \setminus I_0 \cup O_0$, so

$$|\mathcal{Y}_{\times}| = 1 + k(n - k) + \binom{k}{2} + \binom{n - k}{2} - 1 = \binom{n}{2}.$$

On the weight- k slice, pair monomials span all degree-at-most-two functions:

$$x_i = \frac{1}{k-1} \sum_{j \neq i} x_i x_j, \quad 1 = \frac{1}{\binom{k}{2}} \sum_{i < j} x_i x_j.$$

It suffices to show that a pair polynomial

$$F(B) = \sum_{\{u,v\} \subseteq B} c_{uv}$$

vanishing on $\mathcal{Y}_\times$ has all coefficients zero.

Write

$$S = \sum_{\{i,j\} \in \binom{I}{2}} c_{ij}, \quad X_i = \sum_{j \in I \setminus \{i\}} c_{ij}, \quad Y_a = \sum_{i \in I} c_{ia}.$$

Evaluation at I gives $S = 0$ and hence $\sum_i X_i = 0$. Evaluation at $I \setminus \{i\} \cup \{a\}$ gives

$$c_{ia} = Y_a - X_i.$$

Summing over i yields $(k-1)Y_a = 0$, so

$$Y_a = 0, \quad c_{ia} = -X_i. \tag{8}$$

Write $O_0 = \{a_0, a_1\}$ and $z_0 = c_{a_0 a_1}$. Evaluation at $I \setminus \{i, j\} \cup O_0$, using (8), gives

$$c_{ij} = -X_i - X_j - z_0. \tag{9}$$

Summing (9) over $j \neq i$ and using $\sum_j X_j = 0$ gives $X_i = -z_0$ for every i . Summing in i gives $z_0 = 0$, so all inside and cross coefficients vanish. Finally, evaluation at $I \setminus I_0 \cup \{a, b\}$ gives $c_{ab} = 0$ for every outside pair. Thus $F = 0$. $\square$

Theorem 4.19 (Sparse adjacent-two-value Log-Rank from unisolvence). *Assume $2 \leq k \leq n-2$, and let $Y \subseteq \binom{[n]}{k}$ be unisolvent for the degree-at-most-two slice space. Then the following holds simultaneously for every $\mathcal{A} \subseteq 2^{[n]}$ and integer a satisfying*

$$|A \cap B| \in \{a, a+1\} \quad (A \in \mathcal{A}, B \in Y).$$

For $M_{A,B} = |A \cap B| - a$ and $r = \text{rank}_{\mathbb{R}} M$,

$$D(M) \leq \lceil \log_2(r+1) \rceil + 1, \quad \text{rect}(M) \geq \frac{1}{4(r+1)}.$$

Proof. For fixed A , the affine function $f_A(B) = |A \cap B| - a$ has slice degree at most one and is Boolean on Y . The selected set is unisolvent through degree two by hypothesis, so Theorem 4.13 makes f_A Boolean on the complete slice. Every Boolean degree-one function on an interior Johnson slice is constant, a coordinate query, or its complement [FI19b].

Choose r basis rows. Their depth-one trees query at most r coordinates, and every other row is a junta over this pooled set. The proof of Lemma 4.5 uses at most one additional representative. Alice sends its index among at most $r+1$ possibilities and Bob returns one bit, proving the communication bound. Finally,

$$2^{-\lceil \log_2(r+1) \rceil - 1} \geq \frac{1}{4(r+1)},$$

so a largest transcript rectangle gives the stated density. $\square$

Corollary 4.20 (Quadratic-column adjacent-two-value Log-Rank). *For every $2 \leq k \leq n-2$, there exists a fixed family $Y \subseteq \binom{[n]}{k}$ with*

$$|Y| = \binom{n}{2}$$

for which all conclusions of Theorem 4.19 hold. Moreover, a sample from the degree-two projection DPP has this property with probability one.

Proof. Take $Y = \mathcal{Y}_\times$ from Theorem 4.18. The final DPP assertion follows from Theorem 4.15 and (7). $\square$

On balanced layers the column count can be reduced further, from quadratic to linear, by separating Booleanity detection from rank preservation. Put

$$\gamma_{n,k} = \frac{\binom{n-4}{k-2}}{\binom{n}{k}} = \frac{k(k-1)(n-k)(n-k-1)}{n(n-1)(n-2)(n-3)}. \quad (10)$$

Lemma 4.21 (Four-coordinate two-value gap). *For every $A \subseteq [n]$ with $2 \leq |A| \leq n-2$ and every integer a ,*

$$\Pr_{B \in \binom{[n]}{k}} [|A \cap B| \in \{a, a+1\}] \leq 1 - \gamma_{n,k}.$$

Proof. Choose $u, v \in A$ and $p, q \notin A$, all distinct, and partition the k -sets by their trace outside $D = \{u, v, p, q\}$. A fiber whose outside trace has size $k-2$ contains the six two-subsets of D . Their local contributions to $|A \cap B|$ are 0, 1, 2 with multiplicities 1, 4, 1. Any two consecutive integers omit at least one extreme. There are $\binom{n-4}{k-2}$ such fibers, proving the claim. $\square$

The fourth-power parameter in Lemma 4.21 is an artifact of activating one fixed four-coordinate fiber. We next prove a quadratic bound. Put

$$p_{n,k} = \frac{\min\{k, n-k\}}{n}, \quad \eta_{n,k} = \frac{p_{n,k}^2}{16}, \quad \xi_{n,k} = \frac{2}{5}p_{n,k}^2, \quad \theta_{n,k} = \max\{\gamma_{n,k}, \xi_{n,k}\}. \quad (11)$$

Lemma 4.22 (Log-concave two-point escape). *Let Z be integer-valued with a finite, interval-supported, log-concave probability mass function. If $\text{Var}(Z) \geq 1/3$, then for every integer a ,*

$$\Pr[Z \notin \{a, a+1\}] \geq \frac{1}{64}.$$

Proof. Write $\varepsilon = \Pr[Z \notin \{a, a+1\}]$, $u = \Pr[Z = a]$, and $v = \Pr[Z = a+1]$, and suppose that $\varepsilon < 1/64$. If $M = \max\{u, v\}$ and $r = \sqrt{\varepsilon/M}$, then

$$M > 63/128, \quad r^2 < 2/63 < 1/25.$$

Log-concavity and the fact that each first tail atom is at most ε give

$$\frac{\Pr[Z = a+2]}{v} \leq r, \quad \frac{\Pr[Z = a-1]}{u} \leq r,$$

with the zero cases interpreted using interval support. Indeed, when $u > v > 0$, the first inequality follows from

$$\frac{\Pr[Z = a+2]}{v} \leq \min\left\{\frac{v}{u}, \frac{\varepsilon}{v}\right\} \leq \sqrt{\frac{\varepsilon}{u}},$$

and the other order and the left tail are symmetric. Successive mass ratios decrease away from a log-concave mode. Put $t_- = \Pr[Z = a-1]$ and $t_+ = \Pr[Z = a+2]$. Hence for $j \geq 1$ the two tail atoms at distance $j+1/2$ from $a+1/2$ are at most $t_- r^{j-1}$ and $t_+ r^{j-1}$, respectively, while $t_- + t_+ \leq \varepsilon$.

Since $r < 1/5$,

$$\sum_{j \geq 1} (j+1/2)^2 r^{j-1} \leq \frac{1+r}{(1-r)^3} + \frac{1}{(1-r)^2} + \frac{1}{4(1-r)} \leq \frac{135}{32} < \frac{17}{4}.$$

Centering at $a+1/2$ therefore gives

$$\text{Var}(Z) \leq \mathbb{E}(Z - a - 1/2)^2 < \frac{1}{4} + \frac{17}{4}\varepsilon < \frac{81}{256} < \frac{1}{3},$$

a contradiction. $\square$

Theorem 4.23 (Quadratic hypergeometric two-point gap). *For every $A \subseteq [n]$ with $2 \leq |A| \leq n - 2$ and every integer a ,*

$$\Pr_{B \in \binom{[n]}{k}} [|A \cap B| \notin \{a, a+1\}] \geq \eta_{n,k}.$$

Proof. Complementing B replaces k by $n - k$ and $|A \cap B|$ by $|A| - |A \cap B|$. Complementing A replaces $|A|$ by $n - |A|$ and the intersection by $k - |A \cap B|$. Thus assume $2 \leq k, m := |A| \leq n/2$. Set

$$p = k/n, \quad q = m/n, \quad \mu = \mathbb{E}|A \cap B| = pqn.$$

The hypergeometric mass ratios

$$\frac{\Pr[X = j+1]}{\Pr[X = j]} = \frac{(m-j)(k-j)}{(j+1)(n-m-k+j+1)}$$

decrease in j , so the law is interval-supported and log-concave.

Suppose first that $\mu \geq 1$. Since

$$p + q \leq \frac{1}{2} + 2pq$$

for $p, q \leq 1/2$, the exact hypergeometric variance satisfies

$$\text{Var}(X) = \mu(1-p)(1-q) \frac{n}{n-1} \geq \frac{\mu(n-2\mu)}{2(n-1)} \geq \frac{1}{3}.$$

The last quadratic is concave on $[1, n/4]$, and both endpoint values are at least $1/3$. Lemma 4.22 gives escape at least $1/64 \geq p^2/16$.

Now suppose $\mu < 1$. For $a \leq -2$ the conclusion is immediate. For $a = -1$, fixing one element of A gives escape probability at least $p \geq p^2/16$. For $a = 0$, fixing two elements gives

$$\Pr[X \geq 2] \geq \frac{k(k-1)}{n(n-1)} \geq \frac{p^2}{2}.$$

Finally let $a \geq 1$. Then $X = 0$ is an escape value. Integrality of $km < n$ and $2 \leq k \leq n/2$ gives, for $0 \leq i < k$,

$$\frac{m}{n-i} \leq \frac{n-1}{k(n-k+1)} \leq \frac{1}{2}, \quad \sum_{i=0}^{k-1} \frac{m}{n-i} < 2.$$

Here the bound by $1/2$ follows from $k(n-k+1) - 2(n-1) = (k-2)(n-k-1) \geq 0$. By concavity, $\log(1-x) \geq -x \log 4$ on $[0, 1/2]$. Therefore

$$\Pr[X = 0] = \prod_{i=0}^{k-1} \left(1 - \frac{m}{n-i}\right) > \frac{1}{16} \geq \frac{p^2}{16}.$$

This covers every adjacent pair. □

Proposition 4.24 (Sharp $2/5$ gap for row short-side at most three). *Let*

$$q = \min\{k, n-k\}, \quad s = \min\{|A|, n-|A|\}.$$

If $s \in \{2, 3\}$, then for every integer a ,

$$\Pr_{B \in \binom{[n]}{k}} [|A \cap B| \notin \{a, a+1\}] \geq \frac{2}{5} \left(\frac{q}{n}\right)^2.$$

Equality holds if and only if

$$(n, k, |A|, a) = (6, 3, 3, 1).$$

Proof. Complementing A and B reduces to $2 \leq k \leq n/2$ and $m := |A| \in \{2, 3\}$. For $m = 2$, the least mass left outside an adjacent pair is

$$\frac{k(k-1)}{n(n-1)} > \frac{2}{5} \left(\frac{k}{n}\right)^2.$$

Let $m = 3$, write $P_j = \Pr[|A \cap B| = j]$, and put $D = n(n-1)(n-2)$. Then

$$\begin{aligned} P_0 &= \frac{(n-k)(n-k-1)(n-k-2)}{D}, & P_1 &= \frac{3k(n-k)(n-k-1)}{D}, \\ P_2 &= \frac{3k(k-1)(n-k)}{D}, & P_3 &= \frac{k(k-1)(k-2)}{D}. \end{aligned}$$

Since $P_0 \geq P_3$ and $P_1 \geq P_2$, it is enough to bound $P_2 + P_3$ and $P_0 + P_3$.

Write $n = 2k + t$, $t \geq 0$. After clearing positive denominators, the first bound is the strict positivity of

$$\begin{aligned} H(k, t) &= 16k(k-1)(2k-1) \\ &\quad + (42k^2 - 54k + 10)t + (13k - 15)t^2, \end{aligned}$$

whose displayed coefficients are positive for $k \geq 2$. The second bound is equivalent to $F(k, t) \geq 0$, where

$$\begin{aligned} F(k, t) &= 12k^2(k-1)(k-3) + (32k^3 - 84k^2 + 40k)t \\ &\quad + (43k^2 - 60k + 10)t^2 + (25k - 15)t^3 + 5t^4. \end{aligned}$$

For $k \geq 3$, every coefficient is nonnegative; the only possible zero is $k = 3, t = 0$. For $k = 2$, the constraint $m = 3 \leq n/2$ gives $t \geq 2$, and

$$F(2, t) = -48 + 62t^2 + 35t^3 + 5t^4 > 0.$$

Thus equality occurs only at $n = 6, k = 3$, where the escaping mass $P_0 + P_3 = 1/10$ corresponds to the captured pair $\{1, 2\}$, hence $a = 1$. $\square$

Proposition 4.25 (Sharp $2/5$ gap on the low-density side). *If*

$$\frac{\min\{k, n-k\}}{n} \leq \frac{1}{3},$$

then for every nontrivial $A \subseteq [n]$ and every integer a ,

$$\Pr_{B \in \binom{[n]}{k}} [|A \cap B| \notin \{a, a+1\}] > \frac{2}{5} \left(\frac{\min\{k, n-k\}}{n} \right)^2.$$

Proof. Complement B and write $n = 3k + t$, where $k \geq 2$ and $t \geq 0$. After substituting the four-coordinate lower bound and clearing positive denominators, the assertion is equivalent to $L(k, t) > 0$, where

$$\begin{aligned} L(k, t) &= 6k^4 + 18k^3 - 36k^2 + 12k + (26k^3 - 33k^2 + 3k)t \\ &\quad + (17k^2 - 28k + 5)t^2 + (3k - 5)t^3. \end{aligned}$$

Putting $k = q + 2$ gives

$$\begin{aligned} L(q+2, t) &= 6q^4 + 66q^3 + 216q^2 + 276q + 120 \\ &\quad + (26q^3 + 123q^2 + 183q + 82)t \\ &\quad + (17q^2 + 40q + 17)t^2 + (3q + 1)t^3, \end{aligned}$$

which is strictly positive for $q, t \geq 0$. $\square$

Propositions 4.24 and 4.25 reduce the unresolved sharp 2/5 theorem to

$$\frac{1}{3} < \frac{\min\{k, n-k\}}{n} \leq \frac{1}{2}, \quad \min\{|A|, n-|A|\} \geq 4.$$

Theorem 4.26 (Global sharp hypergeometric two-point gap). *For every $A \subseteq [n]$ with $2 \leq |A| \leq n-2$ and every integer a ,*

$$\Pr_{B \in \binom{[n]}{k}} [|A \cap B| \notin \{a, a+1\}] \geq \frac{2}{5} \left(\frac{\min\{k, n-k\}}{n} \right)^2.$$

Equality holds if and only if

$$(n, k, |A|, a) = (6, 3, 3, 1).$$

Proof. Complement A and B so that $2 \leq k \leq n/2$ and $2 \leq m := |A| \leq n/2$. The case $m = 2$ is strict by Proposition 4.24. For $m \geq 3$, fix three points of A and three of its complement. Partition the k -sets by their trace outside these six coordinates.

For local trace sizes 2, 3, 4, the local intersection multiplicities are

$$(3, 9, 3), \quad (1, 9, 9, 1), \quad (3, 9, 3).$$

Every adjacent pair misses at least 3, 2, 3 local columns, respectively. Summing the fibers gives escape probability at least

$$\zeta_{n,k} = \frac{3 \binom{n-6}{k-2} + 2 \binom{n-6}{k-3} + 3 \binom{n-6}{k-4}}{\binom{n}{k}}.$$

Here and below, out-of-range binomial coefficients are interpreted as zero.

Write $n = 2k + t$, $t \geq 0$. Clearing the positive denominators in $\zeta_{n,k} \geq 2(k/n)^2/5$ gives $G(k, t) \geq 0$. For $k = q + 3$,

$$\begin{aligned} G(q+3, t) = & 16q^6 + 228q^5 + 1180q^4 + 2820q^3 + 3124q^2 + 1272q \\ & + (120q^5 + 1140q^4 + 4040q^3 + 6480q^2 + 4432q + 876)t \\ & + (230q^4 + 1665q^3 + 4325q^2 + 4710q + 1750)t^2 \\ & + (195q^3 + 1065q^2 + 1925q + 1140)t^3 \\ & + (80q^2 + 300q + 290)t^4 + (13q + 24)t^5. \end{aligned}$$

All coefficients are nonnegative, and the unique zero is $q = t = 0$. For $k = 2$, one has $t \geq 2$; putting $t = s + 2$ gives

$$G(2, s+2) = 11s^5 + 180s^4 + 1085s^3 + 3000s^2 + 3764s + 1680 > 0.$$

Thus equality requires $n = 6, k = 3$. Then the six-coordinate fiber is the whole ground set, $m = 3$, and its multiplicities are $(1, 9, 9, 1)$. Exactly two of twenty columns escape only when the captured pair is $\{1, 2\}$, giving $a = 1$. $\square$

Combining Lemma 4.21 and Theorem 4.26, every nontrivial row escapes every consecutive pair on at least a $\theta_{n,k}$ -fraction of the layer.

We also need an explicit linear-size rank anchor. Fix $i_0 \in I$ and $a_0 \notin I$, and set

$$\mathcal{Y}_0 = \{I\} \cup \{I \setminus \{i_0\} \cup \{a\} : a \notin I\} \cup \{I \setminus \{i\} \cup \{a_0\} : i \in I \setminus \{i_0\}\}. \quad (12)$$

Then $|\mathcal{Y}_0| = n$.

Lemma 4.27 (Linear degree-one anchor). *The family $\mathcal{Y}_0$ is unisolvent for the degree-at-most-one slice space.*

Proof. If $f(B) = c + \sum_{j \in B} w_j$ vanishes on $\mathcal{Y}_0$, comparison with I gives $w_a = w_{i_0}$ for every $a \notin I$, and $w_i = w_{a_0}$ for every $i \in I \setminus \{i_0\}$. Thus all w_j are equal, so f is constant on the slice; its value at I is zero. $\square$

Theorem 4.28 (Linear-size sparse HLS on balanced layers). *Let $\mathcal{Y}_1$ be a uniformly random N -element subset of $\binom{[n]}{k}$, sampled without replacement, and put $\mathcal{Y} = \mathcal{Y}_0 \cup \mathcal{Y}_1$. If*

$$N \geq \frac{n \log 2 + \log(n+2) + \log(1/\delta)}{\theta_{n,k}},$$

then, with probability at least $1 - \delta$, simultaneously for every $\mathcal{A} \subseteq 2^{[n]}$ and integer a satisfying

$$|A \cap B| \in \{a, a+1\} \quad (A \in \mathcal{A}, B \in \mathcal{Y}),$$

the matrix $M_{A,B} = |A \cap B| - a$ satisfies, with $r = \text{rank}_{\mathbb{R}} M$,

$$D(M) \leq \lceil \log_2(r+1) \rceil + 1, \quad \text{rect}(M) \geq \frac{1}{4(r+1)}.$$

Moreover $|\mathcal{Y}| \leq n + N$.

Proof. For fixed nontrivial A and a , the combined $\theta_{n,k}$ gap and sampling without replacement give failure probability at most $e^{-\theta_{n,k}N}$. A union bound over at most 2^n sets and $n+2$ relevant consecutive pairs shows that, outside probability δ , every admissible A is empty, full, a singleton, or a co-singleton.

Because $\mathcal{Y}$ contains the degree-one-unisolvent anchor $\mathcal{Y}_0$, every row relation lifts to the complete slice and the ambient matrix has the same real rank r . Every coordinate is nonconstant on $\mathcal{Y}_0$; hence the two-value condition forces the corresponding rows to be constants, coordinates, or coordinate complements. The rank-pooled depth-one protocol from Theorem 4.19 gives the communication and rectangle bounds. $\square$

Theorem 4.29 (Deletion-robust linear sparse HLS). *If*

$$N \geq \frac{8}{\theta_{n,k}} (n \log 2 + \log(n+2) + \log(1/\delta)),$$

then, with probability at least $1 - \delta$, every nontrivial A and every consecutive pair $\{a, a+1\}$ have at least

$$\theta_{n,k}N/2$$

witnesses $B \in \mathcal{Y}_1$ with $|A \cap B| \notin \{a, a+1\}$. Consequently all conclusions of Theorem 4.28 remain true after an adversary deletes any $D \subseteq \mathcal{Y}_1$ satisfying

$$|D| \leq \theta_{n,k}N/4,$$

while the anchor $\mathcal{Y}_0$ is retained.

Proof. For fixed A, a , the number of witnesses in the without-replacement sample is hypergeometric with mean at least $\theta_{n,k}N$. Its lower-tail Chernoff bound gives

$$\Pr[X < \theta_{n,k}N/2] \leq e^{-\theta_{n,k}N/8}.$$

Union bound over all A, a . After the stated deletion, at least $\theta_{n,k}N/4$ witnesses remain, and the rank anchor is unchanged. $\square$

We now remove the requirement that the deterministic rank anchor be protected. The first input is an elementary minimum-support estimate for affine functions on a slice. Put

$$\alpha_{n,k} = \frac{\binom{n-2}{k-1}}{\binom{n}{k}} = \frac{k(n-k)}{n(n-1)}.$$

Lemma 4.30 (Affine slice support gap). *If a degree-at-most-one function f is not identically zero on $\binom{[n]}{k}$, then*

$$|\{B \in \binom{[n]}{k} : f(B) \neq 0\}| \geq \binom{n-2}{k-1} = \alpha_{n,k} \binom{n}{k}.$$

Proof. Write $f(B) = c + \sum_{i \in B} w_i$. If all w_i are equal, then f is a nonzero constant on the slice and the claim is immediate. Otherwise choose i, j with $w_i \neq w_j$. For every $U \in \binom{[n] \setminus \{i, j\}}{k-1}$, pair the two columns $U \cup \{i\}$ and $U \cup \{j\}$. Their f -values differ by $w_i - w_j$, so at least one is nonzero. The pairs are disjoint and there are $\binom{n-2}{k-1}$ of them. $\square$

We use the following standard finite-population form of VC uniform convergence. There is an absolute constant C_{VC} such that, for a finite set Ω , a class $\mathcal{R} \subseteq 2^\Omega$ of VC dimension at most d , and a uniform N -element subset Y , the inequalities

$$N \geq C_{\text{VC}} \varepsilon^{-2} (d \log(2/\varepsilon) + \log(2/\eta))$$

imply, with probability at least $1 - \eta$,

$$\sup_{R \in \mathcal{R}} \left| \frac{|R \cap Y|}{N} - \frac{|R|}{|\Omega|} \right| \leq \varepsilon.$$

This is the classical VC uniform-convergence theorem, with the usual without-replacement concentration comparison [VC71, BM15]. For completeness, if W is a d -dimensional function space, then the support ranges $\{\text{supp}(f) : 0 \neq f \in W\}$ have VC dimension at most d : shattering $d+1$ points would produce $d+1$ functions whose restrictions are nonzero multiples of the coordinate basis vectors, contradicting $\dim W = d$.

Theorem 4.31 (Anchor-free fully deletion-robust sparse HLS). *Let $\mathcal{Y}$ be a uniformly random N -element subset of $\binom{[n]}{k}$. If*

$$N \geq \max \left\{ \frac{8}{\theta_{n,k}} (n \log 2 + \log(n+2) + \log(2/\delta)), \frac{4C_{\text{VC}}}{\alpha_{n,k}^2} (n \log(4/\alpha_{n,k}) + \log(4/\delta)) \right\},$$

then, with probability at least $1 - \delta$, the following holds simultaneously for every deletion set

$$D \subseteq \mathcal{Y}, \quad |D| \leq \frac{N}{4} \min\{\theta_{n,k}, \alpha_{n,k}\}.$$

For every $\mathcal{A} \subseteq 2^{[n]}$ and integer a satisfying

$$|A \cap B| \in \{a, a+1\} \quad (A \in \mathcal{A}, B \in \mathcal{Y} \setminus D),$$

the matrix $M_{A,B} = |A \cap B| - a$, restricted to columns $\mathcal{Y} \setminus D$, satisfies, with $r = \text{rank}_{\mathbb{R}} M$,

$$\text{D}(M) \leq \lceil \log_2(r+1) \rceil + 1, \quad \text{rect}(M) \geq \frac{1}{4(r+1)}.$$

No distinguished subset of columns is protected from deletion.

Proof. Apply the preceding VC estimate with error $\alpha_{n,k}/2$ to the support ranges of the n -dimensional degree-at-most-one slice space. By Lemma 4.30, except with probability $\delta/2$, every nonzero affine slice function is nonzero on at least $\alpha_{n,k}N/2$ sampled columns.

The hypergeometric lower-tail argument in Theorem 4.29, now with failure probability $\delta/2$, says that every nontrivial A and consecutive pair have at least $\theta_{n,k}N/2$ witness columns. Both events therefore hold with probability at least $1 - \delta$.

After any allowed deletion, every nonzero affine function remains nonzero on some retained column. Hence $\mathcal{Y} \setminus D$ is degree-one unisolvent. Likewise every nontrivial set A retains a witness outside every consecutive pair, so every admissible row set is empty, full, a singleton, or a co-singleton. Unisolvence lifts all row relations to the complete slice and preserves the actual rank. The rank-pooled depth-one protocol used in Theorem 4.19 finishes the proof. $\square$

If $\beta n \leq k \leq (1 - \beta)n$, then

$$\theta_{n,k} \geq \xi_{n,k} \geq 2\beta^2/5.$$

Thus for fixed $\delta < 1$, Theorem 4.28 proves the deterministic existence of universal sparse HLS families with only $O_\beta(n)$ columns. Theorem 4.29 uses $O(\beta^{-2}(n + \log(1/\delta)))$ detector columns and tolerates deletion of a $\beta^2/10$ -fraction of them. Moreover

$$\alpha_{n,k} \geq \beta(1 - \beta) \geq 2\beta^2/5,$$

so Theorem 4.31 also uses $O(\beta^{-2}(n \log(1/\beta) + \log(1/\delta)))$ columns and tolerates deletion of a $\beta^2/10$ -fraction of the entire column family, with no protected anchor.

4.3 Deterministic spectral and design columns

The preceding existence theorem can be replaced by a deterministic spectral hypothesis. For $B \in Y \subseteq \binom{[n]}{k}$, write $x_B \in \{0, 1\}^n$ for its incidence vector and define

$$\bar{x}_Y = \frac{1}{|Y|} \sum_{B \in Y} x_B, \quad \Sigma_Y = \frac{1}{|Y|} \sum_{B \in Y} (x_B - \bar{x}_Y)(x_B - \bar{x}_Y)^\top.$$

Since every column has weight k , one has $\Sigma_Y \mathbf{1} = 0$.

Theorem 4.32 (Deterministic covariance-gap sparse HLS). *Assume $n \geq 4$, $2 \leq k \leq n - 2$, $Y \neq \emptyset$, and, for every $v \perp \mathbf{1}$,*

$$v^\top \Sigma_Y v \geq \lambda \|v\|_2^2, \quad \lambda > \frac{n}{8(n-2)}. \quad (13)$$

Then the following holds simultaneously for every $\mathcal{A} \subseteq 2^{[n]}$ and integer a satisfying

$$|A \cap B| \in \{a, a+1\} \quad (A \in \mathcal{A}, B \in Y).$$

For $M_{A,B} = |A \cap B| - a$ and $r = \text{rank}_{\mathbb{R}} M$,

$$D(M) \leq \lceil \log_2(r+1) \rceil + 1, \quad \text{rect}(M) \geq \frac{1}{4(r+1)}.$$

In particular, this is a deterministic theorem for every prescribed column family satisfying (13).

Proof. Fix $A \subseteq [n]$, put $m = |A|$, and let

$$v_A = \mathbf{1}_A - \frac{m}{n} \mathbf{1} \in \mathbf{1}^\perp.$$

Because $\Sigma_Y \mathbf{1} = 0$,

$$\text{Var}_{B \in Y} |A \cap B| = \mathbf{1}_A^\top \Sigma_Y \mathbf{1}_A = v_A^\top \Sigma_Y v_A \geq \lambda \frac{m(n-m)}{n}.$$

If $2 \leq m \leq n-2$, then

$$\frac{m(n-m)}{n} \geq \frac{2(n-2)}{n},$$

so (13) makes the variance strictly larger than $1/4$. This contradicts the fact that a random variable supported on $\{a, a+1\}$ has variance at most $1/4$. Hence every admissible A is empty, full, a singleton, or a co-singleton.

The same spectral gap proves degree-one unisolvence without a separate anchor. Indeed, suppose

$$f(B) = c + w^\top x_B$$

vanishes on Y . Writing $w = w_0 + t\mathbf{1}$ with $w_0 \perp \mathbf{1}$, the empirical variance of f is zero, whereas (13) gives

$$0 = w_0^\top \Sigma_Y w_0 \geq \lambda \|w_0\|_2^2.$$

Thus $w_0 = 0$, and f is constant on the complete weight- k slice. Its value on the nonempty set Y is zero, so it vanishes on the complete slice.

Consequently every row relation on Y lifts to the complete slice and the ambient and restricted matrices have the same real rank. The positive gap also makes every coordinate nonconstant on Y . Hence the two-value condition forces singleton rows to be coordinates, co-singleton rows to be coordinate complements, and the remaining classified rows to be constants, all with Boolean extensions to the complete slice. The rank-pooled depth-one protocol from Theorem 4.19 gives the communication bound, and a largest transcript rectangle gives the stated density. $\square$

A large, standard source of deterministic spectral families is furnished by block designs.

Corollary 4.33 (2-design columns). *Let Y be the block family of a simple 2 -(n, k, λ_2) design. If*

$$8k(n-k)(n-2) > n^2(n-1), \quad (14)$$

then all conclusions of Theorem 4.32 hold.

Proof. Write $b = |Y|$ and let r_2 be the number of blocks through one point. The design identities give

$$\frac{r_2}{b} = \frac{k}{n}, \quad \frac{\lambda_2}{b} = \frac{k(k-1)}{n(n-1)}.$$

It follows entrywise that

$$\Sigma_Y = \frac{k(n-k)}{n(n-1)} \left(I - \frac{1}{n} J \right).$$

Thus the eigenvalue on $\mathbf{1}^\perp$ is $k(n-k)/[n(n-1)]$, and (14) is exactly the strict threshold in (13). $\square$

The design corollary yields an explicit information-theoretically minimal family on an infinite sequence of central layers.

Corollary 4.34 (Explicit Sylvester–Hadamard columns). *For every $t \geq 3$, put*

$$n = 2^t - 1, \quad k = 2^{t-1} - 1.$$

There is an explicit family

$$Y_t \subseteq \binom{[n]}{k}, \quad |Y_t| = n,$$

for which all conclusions of Theorem 4.32 hold simultaneously for every adjacent-two-value row family. The column count n is minimal for degree-one unisolvence.

Proof. Let H_t be the Sylvester Hadamard matrix, recursively defined by

$$H_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H_{t+1} = \begin{pmatrix} H_t & H_t \\ H_t & -H_t \end{pmatrix}.$$

It has order $4u = 2^t$, where $u = 2^{t-2}$, and its first row and column are all $+1$. Delete that row and column. For every remaining row, take the set of positions carrying $+1$. Orthogonality shows that every block has size $2u - 1$, and every two blocks meet in exactly $u - 1$ points. Hence these $4u - 1 = n$ blocks form a symmetric 2 -($4u - 1, 2u - 1, u - 1$) design.

Its covariance eigenvalue on $\mathbf{1}^\perp$ is

$$\frac{k(n-k)}{n(n-1)} = \frac{u}{4u-1}.$$

For $u \geq 2$,

$$\frac{u}{4u-1} > \frac{4u-1}{8(4u-3)}$$

because the difference after clearing positive denominators is $16u^2 - 16u - 1 > 0$. Theorem 4.32 therefore applies. Finally, the degree-at-most-one slice space has dimension n , so an unisolvent family cannot have fewer than n columns. $\square$

There is an analogous minimal sparse family for every bounded-alphabet complete product. For $\Omega_{\text{prod}} = \prod_i [m_i]$, decompose each one-coordinate function space into constants and its $(m_i - 1)$ -dimensional mean-zero part. The degree-at-most- s product space then has dimension

$$d_s^{\text{prod}} = \sum_{\substack{J \subseteq [n] \\ |J| \leq s}} \prod_{i \in J} (m_i - 1). \quad (15)$$

It is invariant under the transitive action of $\prod_i S_{m_i}$.

Corollary 4.35 (Minimal sparse bounded-product columns). *Sample Y from the rank- d_s^{prod} projection DPP of the degree-at-most- s space on Ω_{prod} . With probability one,*

$$|Y| = d_s^{\text{prod}}$$

and every Boolean degree- s row family restricted to Y satisfies the communication bound in Theorem 4.10, with its actual restricted real rank.

Proof. Apply Theorems 4.15 and 4.12, using (15). $\square$

For fixed Q, s ,

$$d_s^{\text{prod}} \leq \sum_{j=0}^s \binom{n}{j} (Q-1)^j = O_{Q,s}(n^s),$$

whereas $|\Omega_{\text{prod}}| \geq 2^n$. Thus this construction is again information-theoretically minimal and exponentially sparse.

For fixed s , the uniform Johnson construction uses $O_s(n^s \log n)$ columns. A middle Johnson layer has $\Theta(2^n / \sqrt{n})$ columns, so the retained family is exponentially sparse. Taking any fixed $\delta < 1$ also proves deterministic existence of such sparse families. More basically, every d -dimensional function space has some unisolvent set of exactly d points, by choosing a basis of its evaluation functionals. Projection-DPP sampling attains this minimum exactly, while the uniform theorem gives an elementary without-replacement construction at logarithmic oversampling.

5 The product of Johnson middle layers

Let

$$[d] = C_1 \sqcup \cdots \sqcup C_m, \quad |C_i| = n,$$

where $n \geq 4$ is even, and define

$$\mathcal{Y}_{n,m} = \{B \subseteq [d] : |B \cap C_i| = n/2 \text{ for all } i\}. \quad (16)$$

Theorem 5.1 (Product-Johnson restricted Log-Rank theorem). *Let $\mathcal{A} \subseteq 2^{[d]}$ be nonempty and suppose*

$$|A \cap B| \in \{a, a+1\} \quad (A \in \mathcal{A}, B \in \mathcal{Y}_{n,m}).$$

Then the matrix (2) satisfies

$$\text{rank}_{\mathbb{R}} M \leq d+1, \quad \text{rect}(M) \geq \frac{1}{4(d+1)}, \quad D(M) \leq \lceil \log_2(\text{rank}_{\mathbb{R}} M + 1) \rceil + 4.$$

Proof. Choose B uniformly from $\mathcal{Y}_{n,m}$. The restrictions $B \cap C_i$ are independent uniform middle-layer sets. Put $s_i = |A \cap C_i|$. The hypergeometric variance formula gives

$$\text{Var } |A \cap B| = \sum_{i=1}^m \frac{s_i(n-s_i)}{4(n-1)}. \quad (17)$$

By Lemma 2.1, the left side is at most $1/4$.

Let

$$h(A) = \sum_i \min\{s_i, n-s_i\},$$

the Hamming distance from A to the nearest union of blocks. If $h(A) \geq 2$, then either two errors occur in one block or errors occur in at least two blocks. In the first case the corresponding numerator in (17) is at least $2(n-2)$; in the second it is at least $2(n-1)$. Here we use that $r(n-r)$ is increasing for $0 \leq r \leq n/2$, so additional error units can only increase the relevant lower bound. Since $n \geq 4$, both cases make (17) strictly larger than $1/4$, a contradiction. Therefore $h(A) \leq 1$.

Write

$$A = C_J \triangle E, \quad C_J = \bigcup_{i \in J} C_i, \quad E \in \{\emptyset, \{1\}, \dots, \{d\}\}.$$

Pigeonhole the rows by E , retaining at least a $1/(d+1)$ fraction. If $E = \emptyset$, every retained row has constant intersection with every column because each block count is fixed. If $E = \{e\}$, additionally fix whether $e \in B$; exactly half the columns lie in each class. Every retained row is now constant on the retained columns. Keep the larger of the rows with constant value a and those with constant value $a+1$. The total loss is at most $4(d+1)$.

The rank assertion follows from (3). $\square$

5.1 A rank-sensitive one-way protocol

The row classification in the proof gives a full protocol, not only one large rectangle. If $E = \emptyset$, the row is constant. If $E = \{e\}$ and the block containing e is not in J , then

$$|A \cap B| = |J|n/2 + \mathbf{1}_{\{e \in B\}},$$

so the nonconstant matrix row is exactly the coordinate function $B \mapsto \mathbf{1}_{\{e \in B\}}$. If that block is in J , then

$$|A \cap B| = |J|n/2 - \mathbf{1}_{\{e \in B\}},$$

and the nonconstant row is the complement of the coordinate function.

Let $S \subseteq [d]$ be the coordinates occurring in nonconstant rows and put $k = |S|$. Alice can send one of two constant messages, or the index of $e \in S$ together with the complement bit. Thus there is a one-way protocol with at most $2k + 2$ messages.

We next relate k to the actual real rank $r = \text{rank}_{\mathbb{R}} M$. Let X_S be the matrix whose rows are the coordinate functions $B \mapsto \mathbf{1}_{\{e \in B\}}$, $e \in S$. If f blocks are fully contained in S , then

$$\text{rank } X_S = \begin{cases} k, & f = 0, \\ k - f + 1, & f > 0. \end{cases} \quad (18)$$

To see this, vary one middle-layer block while fixing all other blocks. A linear combination of coordinate functions is constant on that block only when all its coefficients are equal, by exchanging one selected and one unselected coordinate. A partially represented block therefore contributes no relation. Each fully represented block contributes its fixed-sum relation, and the f block sums span one common constant direction.

Since $f \leq k/n$,

$$\text{rank } X_S \geq (1 - 1/n)k. \quad (19)$$

Choose one nonconstant row of M for every $e \in S$. The resulting matrix is obtained from X_S by invertible row sign changes and a rank-one constant perturbation. Hence

$$r \geq \text{rank } X_S - 1 \geq (1 - 1/n)k - 1,$$

and, because $n \geq 4$,

$$k \leq \frac{n}{n-1}(r+1) \leq \frac{4}{3}(r+1). \quad (20)$$

Encoding the $2k + 2$ possible one-way messages costs at most $\lceil \log_2(r+1) \rceil + 3$ bits, which implies the stated bound.

Corollary 5.2 (Optimality of the logarithmic order). *For fixed even $n \geq 4$ and growing m , the order $D(M) = \Theta(\log \text{rank } M)$ in Theorem 5.1 is best possible.*

Proof. Take $\mathcal{A} = \{\{e\} : e \in [d]\}$ and $a = 0$. Then M is the coordinate incidence matrix of $\mathcal{Y}_{n,m}$. Formula (18) with all m blocks fully represented gives

$$\text{rank}_{\mathbb{R}} M = d - m + 1.$$

Every deterministic c -bit protocol partitions a Boolean matrix into at most 2^c monochromatic rectangles, each of real rank at most one. Hence $c \geq \log_2 \text{rank } M$. Theorem 5.1 supplies the matching upper bound up to an additive constant. $\square$

The same proof gives a parameterized theorem for arbitrary products of Johnson layers. Let the blocks have sizes $n_i \geq 2$, choose $1 \leq k_i \leq n_i - 1$, and put

$$\mathcal{Y}(\mathbf{n}, \mathbf{k}) = \{B \subseteq [d] : |B \cap C_i| = k_i \text{ for all } i\}.$$

For $0 \leq s_i \leq n_i$, define

$$V(\mathbf{s}) = \sum_i \frac{s_i(n_i - s_i)k_i(n_i - k_i)}{n_i^2(n_i - 1)}, \quad h(\mathbf{s}) = \sum_i \min\{s_i, n_i - s_i\},$$

and the exact weighted tube radius

$$t(\mathbf{n}, \mathbf{k}) = \max\{h(\mathbf{s}) : V(\mathbf{s}) \leq 1/4\}. \quad (21)$$

Theorem 5.3 (General product-Johnson theorem). *If (1) holds with $\mathcal{B} = \mathcal{Y}(\mathbf{n}, \mathbf{k})$ and $t = t(\mathbf{n}, \mathbf{k})$, then*

$$\text{rank}_{\mathbb{R}} M \leq d + 1, \quad \text{rect}(M) \geq \frac{1}{2^{t+1} \sum_{j=0}^t \binom{d}{j}}.$$

Proof. The block restrictions are independent and the hypergeometric variance is exactly $V(\mathbf{s})$. Lemma 2.1 therefore forces $h(\mathbf{s}) \leq t$. Write each row as a block union symmetric-differenced with an error set of size at most t . Pigeonhole the error set, then fix its exact trace on the right; this costs at most $2^t \sum_{j \leq t} \binom{d}{j}$. Every retained row is now constant across the retained columns, and a final two-colour pigeonhole completes the proof. $\square$

The parameterized theorem also gives a communication protocol.

Theorem 5.4 (Protocol for general product layers). *Let $r = \text{rank}_{\mathbb{R}} M$ and $t = t(\mathbf{n}, \mathbf{k})$. If $t = 0$, then M has only constant rows. For every $t \geq 1$,*

$$D(M) = O(t \log(r + 2) + \log(t + 2)).$$

In particular, if $t \leq (\log(r + 2))^C$, then the Log-Rank Conjecture holds for this product-layer matrix.

Proof. Every row has a sparse affine representation on the column family,

$$M_A(B) = q_A + \sum_{e \in E_A} \sigma_{A,e} \mathbf{1}_{\{e \in B\}}, \quad |E_A| \leq t, \quad (22)$$

where $\sigma_{A,e} \in \{-1, 1\}$ and $q_A \in \mathbb{Z}$. The two-level condition forces $q_A \in [-t, t + 1]$.

Choose r matrix rows spanning the row space, and let S_0 be the union of their error supports. Then $|S_0| \leq rt$. We claim that the union S of the error supports of all rows satisfies

$$|S| \leq 2|S_0| \leq 2rt. \quad (23)$$

To prove the claim, consider the evaluation map from affine coordinate forms to functions on $\mathcal{Y}(\mathbf{n}, \mathbf{k})$. Its kernel consists exactly of forms whose coefficients are constant on each block, with the constant term chosen to cancel $\sum_i \alpha_i k_i$. Indeed, varying a single block and exchanging one selected and one unselected coordinate forces all coefficients in that block to be equal.

For an arbitrary row, subtract a linear combination of the chosen basis-row representations. The difference lies in this kernel. If a block contains a new error coordinate outside S_0 , its kernel coefficient is nonzero; therefore every coordinate of that block outside S_0 must belong to the row's error set. Use the canonical nearest-block-union representation, so every block contributes at most half of its coordinates to an error set. Consequently such a block has at most as many coordinates outside S_0 as inside S_0 . Summing over the disjoint blocks that contain new coordinates proves (23).

Alice now sends q_A , the subset $E_A \subseteq S$, and its signs. The number of possible messages is at most

$$(2t + 2) \sum_{j=0}^t \binom{|S|}{j} 2^j.$$

Using $|S| \leq 2rt$ and the standard binomial-sum estimate gives the claimed communication bound. Bob evaluates (22) from his column. $\square$

5.2 All interior product layers

The strongest product theorem does not require a variance-radius estimate.

Theorem 5.5 (Interior product-Johnson Log-Rank theorem). *Assume*

$$2 \leq k_i \leq n_i - 2 \quad \text{for every block } i.$$

If (1) holds with $\mathcal{B} = \mathcal{Y}(\mathbf{n}, \mathbf{k})$, then

$$\text{rect}(M) \geq \frac{1}{4(d+1)}, \quad D(M) \leq \lceil \log_2(\text{rank}_{\mathbb{R}} M + 1) \rceil + 4.$$

The logarithmic order is optimal up to an additive constant.

Proof. Fix a row A and put $s_i = |A \cap C_i|$. As B_i ranges over all k_i -subsets of C_i , the possible values of $|A \cap B_i|$ are all integers in the interval

$$[\max\{0, k_i - (n_i - s_i)\}, \min\{k_i, s_i\}]. \quad (24)$$

The choices in distinct blocks are independent, so the range of $|A \cap B|$ is the sum of these integer intervals. Since the total range has size at most two, the sum of their widths is at most one.

For $2 \leq k_i \leq n_i - 2$, the interval in (24) has width zero exactly when $s_i \in \{0, n_i\}$, and has width one only when $s_i \in \{1, n_i - 1\}$. Indeed, if $2 \leq s_i \leq n_i - 2$ and $s_i + k_i \leq n_i$, its lower endpoint is zero and its upper endpoint is at least two; if $s_i + k_i > n_i$, its width is $n_i - \max\{s_i, k_i\} \geq 2$.

Consequently all blocks are empty or full in A , except possibly one block where A differs from empty or full in one coordinate. Thus every row is constant, a coordinate function, or its complement. The rectangle proof of Theorem 5.1 applies verbatim.

For the rank-sensitive protocol, let S be the set of queried coordinates. The coordinate-rank calculation (18) remains valid for unequal blocks. Since every block has size at least four, the number of fully represented blocks is at most $|S|/4$, and hence $\text{rank } X_S \geq 3|S|/4$. The rank-one perturbation argument then gives $|S| \leq 4(r+1)/3$, completing the protocol proof. The singleton-row example again proves optimality. $\square$

In the language of Theorem 3.1, interior Johnson factors have atom-growth exponent $\alpha = 1$ and constant $C \leq 4/3$: the centered rank of k queried coordinates is k unless the entire block is queried, in which case it is $k - 1 \geq 3k/4$.

Proposition 5.6 (Sharp boundary obstruction). *If blocks of weight one are allowed, the class contains every Boolean matrix with the corresponding number of columns.*

Proof. Take one block C of size N with $k = 1$, so columns are the singletons $\{j\}$, $j \in C$. Given any Boolean matrix with these N columns, identify each row with the subset $A \subseteq C$ of its one-entries. Then

$$|A \cap \{j\}| \in \{0, 1\}$$

and the resulting adjacent-two-value matrix is the original matrix. The case $k = N - 1$ follows by complementation. $\square$

6 Products of interior multislices

The product/rank argument is not specific to two-color slices. For a vector of positive integers $\kappa = (\kappa_1, \dots, \kappa_\ell)$ with sum N , define the multislice

$$\Omega(\kappa) = \{x \in [\ell]^N : |\{p : x_p = c\}| = \kappa_c \text{ for every } c\}.$$

A function has degree at most one if it lies in the affine span of the coordinate-color indicators $1_{\{x_p=c\}}$.

Consider a product of multislices $\Omega = \prod_{i=1}^m \Omega(\kappa^{(i)})$, each with alphabet size at most Q . A Boolean matrix with column set Ω is called row-degree-one if every row is a Boolean degree-one function on this product.

Theorem 6.1 (Interior multislice product theorem). *Assume every color multiplicity in every factor is at least two. If M is a row-degree-one Boolean matrix on Ω , then*

$$D(M) \leq \lceil \log_2(\text{rank}_{\mathbb{R}} M + 1) \rceil + Q + 4.$$

For fixed alphabet bound Q , this is optimal-order Log-Rank.

Proof. Filmus–Ihringer show that a Boolean degree-one function on an interior multislice depends only on the color of one position [FI19b, Lemma 8.1]. A Boolean degree-one function on a product is a sum of one-factor functions. Two nonconstant factors would produce at least three sums, so every nonconstant row depends on one position in one factor and has the form

$$x \mapsto 1_{\{x_p \in J\}}$$

or its complement, for a nontrivial color subset J . Up to complement there are at most $2^{Q-1} - 1$ atoms per position.

Let P be the number of positions used by the matrix rows. We claim

$$P \leq 2(\text{rank}_{\mathbb{R}} M + 1). \quad (25)$$

Choose one atom for each used position. Within one multislice factor, an additive coordinate form that vanishes identically has the form $g_p(c) = a_p + h_c$, with its constants satisfying the single histogram relation. This follows by exchanging the colors at two positions. Hence a proper subset of positions contributes independent nonconstant directions; if all N positions are used, the centered rank is at least $N - \ell \geq N/2$, since every color multiplicity is at least two. Across different factors the centered function spaces form a direct sum. Therefore the centered rank of one selected atom per used position is at least $P/2$.

Replacing complements by positive atoms enlarges the row span by at most the constant function, so (25) follows. The total number of used atoms is at most $2^{Q-1}P \leq 2^Q(r+1)$. Alice sends the atom index and complement bit, proving the stated protocol bound. $\square$

Thus interior multislice factors satisfy the atomic growth hypothesis with $\alpha = 1$ and $C \leq 2^Q$.

Proposition 6.2 (Sharp multislice boundary). *If a color of multiplicity one is allowed, row-degree-one matrices on a single multislice include every Boolean matrix with N columns.*

Proof. Use the binary multislice with histogram $(1, N-1)$. A column is identified by the unique position carrying the rare color. For any subset $J \subseteq [N]$, the function

$$x \mapsto \sum_{p \in J} 1_{\{x_p = \text{rare}\}}$$

is Boolean and has degree one; these functions realize arbitrary row supports. $\square$

7 Sparsity and separation from affine sections

The family (16) lies in the global middle layer and has relative density

$$\frac{|\mathcal{Y}_{n,m}|}{\binom{d}{d/2}} = \frac{\binom{n}{n/2}^m}{\binom{mn}{mn/2}}. \quad (26)$$

For fixed n and $m \rightarrow \infty$, Stirling's formula shows that (26) is $\exp(-\Theta_n(d))$. For $n = 4$ it is

$$\Theta(\sqrt{m}(3/8)^m).$$

The family is also generally not a complete affine-code section. For two blocks of size four, label each block by $\{1, 2, 3, 4\}$. In the first block take

$$\{1, 2\}, \quad \{1, 3\}, \quad \{2, 3\},$$

whose XOR is empty. In the second take

$$\{1, 2\}, \quad \{1, 3\}, \quad \{1, 4\},$$

whose XOR is the full block. Pairing corresponding choices gives three product columns with triple XOR of block weights $(0, 4)$. This XOR is still a global middle-layer set but is not in the product family. Any binary affine coset containing the three columns contains their triple XOR, so its intersection with the middle layer cannot equal $\mathcal{V}_{4,2}$.

8 Boolean tubes around a defect space

Let $D \subseteq \mathbb{R}^d$. Define its Boolean templates

$$\mathcal{C}(D) = \{C \subseteq [d] : \mathbf{1}_C \in D\}$$

and the Hamming distance to that family

$$h_D(A) = \min_{C \in \mathcal{C}(D)} |A \Delta C|.$$

For $\lambda > 0$, define the exact Boolean tube radius

$$\tau_\lambda(D) = \max \left\{ h_D(A) : \text{dist}(\mathbf{1}_A, D)^2 \leq \frac{1}{4\lambda} \right\}. \quad (27)$$

The maximum exists because the Boolean cube is finite and $\emptyset$ is a template.

A convenient sufficient estimate is a Boolean Hamming margin

$$\text{dist}(\mathbf{1}_A, D)^2 \geq \gamma h_D(A), \quad (28)$$

which implies

$$\tau_\lambda(D) \leq \lfloor (4\lambda\gamma)^{-1} \rfloor.$$

The exact radius is often sharper than the global margin. For the span of one all-ones vector it recovers the distinction between singleton and two-coordinate errors that is lost in a uniform factor-1/2 estimate.

9 Integer generator entropy

Suppose nonzero $z_1, \dots, z_s \in \mathbb{Z}^d$ span D over $\mathbb{R}$. Put

$$e_i = \|z_i\|_2^2, \quad h_i = \|z_i\|_1.$$

For $u \geq 0$ and an integer $h \geq 1$, define

$$\phi(u, h) = \begin{cases} H_2(4u) + 4u \log h, & u \leq 1/8, \\ \log(h+1), & u > 1/8, \end{cases}$$

where H_2 is binary entropy in natural units.

Lemma 9.1 (Integer entropy envelope). *Let Z be integer-valued, supported on at most $h + 1$ consecutive integer values, and suppose $\text{Var } Z \leq u$. Then $H(Z) \leq \phi(u, h)$.*

Proof. Let p be the largest atom. For an independent copy Z' , the identity

$$\text{Var } Z = \frac{1}{2} \mathbb{E}(Z - Z')^2$$

shows that $\text{Var } Z \geq p(1 - p)$. Moreover, if $p \leq 1/2$, then

$$\text{Var } Z \geq \frac{1}{2} \Pr[Z \neq Z'] = \frac{1}{2} \left(1 - \sum_x \Pr[Z = x]^2 \right) \geq 1/4.$$

Consequently, when $u \leq 1/8$, one has $p > 1/2$ and $1 - p \leq 2u \leq 4u$. Entropy is maximized by spreading the remaining mass uniformly on the other h support points, giving $H(Z) \leq H_2(4u) + 4u \log h$. For $u > 1/8$, use $H(Z) \leq \log(h + 1)$. $\square$

Let $X = \mathbf{1}_B$ for a uniform random column $B \in \mathcal{B}$, and suppose

$$v^\top \Sigma v \leq \kappa \|v\|_2^2 \quad (v \in D), \quad (29)$$

where $\Sigma = \text{Cov}(X)$. Define

$$\Psi_Z = \sum_{i=1}^s \phi(\kappa e_i, h_i). \quad (30)$$

The integer statistics $T_i = z_i \cdot X$ have variance at most κe_i and range size at most $h_i + 1$. Entropy subadditivity gives

$$\max_t \Pr[(T_1, \dots, T_s) = t] \geq e^{-\Psi_Z}. \quad (31)$$

10 The spectral–lattice rectangle theorem

Theorem 10.1 (Spectral–lattice rectangle theorem). *Let $\mathcal{A}, \mathcal{B} \subseteq 2^{[d]}$ be nonempty and satisfy (1). Let $D \subseteq \mathbb{R}^d$ be spanned by nonzero integer vectors $z_1, \dots, z_s$. Suppose*

$$\Sigma \succeq \lambda P_{D^\perp} \quad (32)$$

and (29). Put $t = \tau_\lambda(D)$ and $S(d, t) = \sum_{j=0}^t \binom{d}{j}$. Then

$$\text{rect}(M) \geq \frac{e^{-\Psi_Z}}{2^{t+1} S(d, t)}. \quad (33)$$

Proof. For fixed $A \in \mathcal{A}$, Lemma 2.1 and (32) give

$$\lambda \text{dist}(\mathbf{1}_A, D)^2 \leq \mathbf{1}_A^\top \Sigma \mathbf{1}_A = \text{Var } |A \cap B| \leq 1/4.$$

By the definition of t , choose $C(A) \in \mathcal{C}(D)$ and $E(A)$ with

$$A = C(A) \triangle E(A), \quad |E(A)| \leq t.$$

There are at most $S(d, t)$ error sets. Retain a largest error fiber.

For the fixed error E , refine $\mathcal{B}$ to a largest atom of

$$(z_1 \cdot \mathbf{1}_B, \dots, z_s \cdot \mathbf{1}_B, B \cap E).$$

By (31), this retains at least $e^{-\Psi_Z} 2^{-t}$ of the columns. Since every template is a real linear combination of the generators, the generator statistics fix $|C(A) \cap B|$. The exact trace on E and the identity

$$|(C \triangle E) \cap B| + 2|C \cap E \cap B| = |C \cap B| + |E \cap B|$$

then fix $|A \cap B|$ for every retained row. Retain the larger row colour, losing at most two. Multiplying the losses gives (33). $\square$

Remark 10.2 (Dimension-free character). Theorem 10.1 has no explicit factor exponential in $\dim D$ or in $|\mathcal{C}(D)|$. Arithmetic complexity is paid through the entropy of the generator statistics.

If $E_2 = \sum_i \|z_i\|_2^2$, $H = \max(1, \max_i \|z_i\|_1)$, and $0 < \kappa \leq 1$, then

$$\Psi_Z \leq O\left(\kappa E_2 \log \frac{e(H+1)}{\kappa}\right), \quad (34)$$

with the convention that the right side is zero for $\kappa = 0$.

Indeed, put $u_i = \kappa e_i$ and $U = \sum_i u_i = \kappa E_2$. For $u_i \leq 1/8$, the bound $H_2(x) \leq x \log(e/x)$ gives

$$\phi(u_i, h_i) \leq 4u_i \log \frac{eH}{u_i}.$$

Since the number of nonzero integer generators is at most E_2 , concavity of $x \mapsto x \log(eH/x)$ bounds the sum of these terms by $O(U \log(eH/\kappa))$. There are at most $8U$ indices with $u_i > 1/8$, and each contributes at most $\log(H+1)$. This proves (34).

11 Robust transfer from a nearby rational subspace

The exact arithmetic hypothesis admits a perturbative extension. Let D_0 be the true spectral defect space and $D_{\mathbb{Q}}$ an integer-generated subspace, with projections $P_0, P_{\mathbb{Q}}$ satisfying

$$\|P_0 - P_{\mathbb{Q}}\|_{\text{op}} \leq \varepsilon. \quad (35)$$

Assume D_0 has outside gap λ and Boolean margin γ_0 , while $D_{\mathbb{Q}}$ has Boolean margin $\gamma_{\mathbb{Q}}$. Suppose covariance on D_0 is at most κ .

Every Boolean template in D_0 lies within Euclidean distance $\varepsilon\sqrt{d}$ of $D_{\mathbb{Q}}$, hence within Hamming distance

$$s = \left\lfloor \frac{\varepsilon^2 d}{\gamma_{\mathbb{Q}}} \right\rfloor \quad (36)$$

of an exact Boolean template there. Moreover, since $\|\Sigma\|_{\text{op}} \leq \text{tr } \Sigma \leq d/4$, covariance on $D_{\mathbb{Q}}$ is at most

$$\kappa_{\mathbb{Q}} = 2\kappa + \frac{d}{2}\varepsilon^2. \quad (37)$$

Theorem 11.1 (Robust rational transfer). *Under the preceding assumptions, let*

$$t = \lfloor (4\lambda\gamma_0)^{-1} \rfloor, \quad T = t + s.$$

If z_i are integer generators for $D_{\mathbb{Q}}$ and $\Psi_{\mathbb{Q}}$ is defined using $\kappa_{\mathbb{Q}}$, then

$$\text{rect}(M) \geq \frac{e^{-\Psi_{\mathbb{Q}}}}{2^{T+1} \sum_{j=0}^T \binom{d}{j}}.$$

Proof. The outside gap and margin put every row within t coordinates of a Boolean template in D_0 . Projector closeness and the rational margin put that template within s coordinates of one in $D_{\mathbb{Q}}$. The Hamming triangle inequality gives total error at most T . The covariance estimate (37) permits the extraction argument of Theorem 10.1 on the rational space with the enlarged error budget; the outside-gap step has already been supplied by D_0 . $\square$

For an orthonormal basis $w_1, \dots, w_q$ of D_0 with rational approximants $r_i = z_i/M_i$ satisfying $\|r_i - w_i\|_2 \leq \eta$ and $M_i \leq H_0$, one has, when $\eta\sqrt{q} < 1$,

$$\|P_0 - P_{\mathbb{Q}}\|_{\text{op}} \leq \frac{2\eta\sqrt{q}}{1 - \eta\sqrt{q}}, \quad \sum_i \|z_i\|_2^2 \leq qH_0^2(1 + \eta)^2.$$

Thus the robust theorem gives an explicit endpoint for simultaneous inverse Littlewood–Offord basis approximation.

12 Relation to previous work

The cross-intersection formulation is due to Hambardzumyan, Lovett, and Shirley [HLS25]. Their theorem is an equivalence, not a solution for the product family considered here. The general $2^{-O(\sqrt{r})}$ rectangle bound of Sudakov–Tomon applies, but is exponentially weaker than Theorem 5.1 when $r \asymp d$ [ST25].

Kupavskii–Weltge prove $|\mathcal{A}||\mathcal{B}| \leq (d+1)2^d$ for full-span binary scalar products, and Kupavskii–Tsarev prove sharp near-extremal stability [KW22, KT26]. Those are absolute cardinality statements. They do not yield a relative monochromatic rectangle inside a given exponentially sparse product family.

Rao–Yehudayoff prove powerful anti-concentration results for large cube families and certain finite-field structured distributions [RY22]. Nguyen proves an inverse Littlewood–Offord theorem for the complete half layer [Ngu13]. These results concern ambient concentration and do not directly give Theorem 10.1 for an arbitrary sparse right family.

There is also a substantial Erdős–Ko–Rado literature for direct products of uniform families. For example, Yao–Lv–Wang determine extremal t -intersecting subfamilies in a multipart setting [YLW23]. Those results impose the one-family condition $|A \cap A'| \geq t$. They do not treat a cross pair with a prescribed full product on one side, two adjacent intersection values, or the resulting rank-sensitive communication protocol.

The one-block local classification in Theorem 5.5 is consistent with the Boolean degree-one theorem of Filmus–Ihringer [FI19b]. Their paper does not state the product communication theorem, the rank-sensitive support argument, or the weight-one universality boundary. Accordingly, novelty here rests on those global and communication-complexity components, not on re-claiming the single-scheme degree-one classification.

Likewise, Filmus–Ihringer’s multislice classification [FI19b, Lemma 8.1] is the local input to Theorem 6.1. The product assembly, centered-rank estimate, rank-sensitive protocol, and count-one universality boundary are the claims made here.

The low-degree slice theorems use published results of Filmus–Ihringer, Chiarelli–Hatami–Saks, Midrijanis, Filmus, and Dafni–Filmus–Lifshitz–Lindzey–Vinyals [FI19a, CHS20, Mid04, Fil22, DFL⁺21]. In particular, the last authors already prove polynomial relations between degree and query complexity on slices. None of these inputs is claimed as new. The contribution here is the row-space support pooling, synchronization of non-unique slice supports, symmetric-tree canonicalization, and the resulting rank-sensitive communication theorems, including the bounded-alphabet product and multislice extensions. The remaining quantitative gap is between the sharp $2s$ threshold with unspecified dependence, the polynomial $144s^4$ query regime, and the sharper s -dependence in the C^s cube-extension regime.

Concept-class evaluation is used by Feldman–Xiao to relate randomized one-way communication to differentially private learning [FX14]. Mande–Sanyal–Sherif relate one-way communication and non-adaptive decision trees for functions composed with fixed gadgets [MSS21]. These works do not state a deterministic real-rank-sensitive bounded-answer evaluation theorem. Theorem 4.6 instead applies directly to an arbitrary Boolean evaluation matrix and replaces the ambient coordinate cost by its real row rank. This targeted comparison is still not a substitute for an independent priority search in learning theory.

The sparse-column extension uses standard constant-leverage sampling on a transitive invariant subspace, matrix Chernoff, and the Gross–Nesme without-replacement comparison [Tro12, GN10]. Those concentration ingredients are not new. The claim here is the unisolvent rank-lifting step and its combination with Theorem 4.6, yielding one sparse column family that works simultaneously for all eligible Boolean row families. The twice-degree certificate $f(f-1)$ additionally removes the need to assume that rows are Boolean outside the sparse family; this is the step used in Theorem 4.19.

Amireddy–Behera–Srinivasan–Sudan–Willumsgaard recently construct constant-query ran-

domized low-degree tests on balanced Boolean slices [ABS⁺26]. Their goal is to distinguish low degree from functions far from low degree using fresh random oracle queries. Theorems 4.13 and 4.19 instead use one fixed minimal interpolation set, certify exact Booleanity for every affine row simultaneously, and derive a deterministic real-rank communication protocol.

Maximal rank and diagonal forms for the full pair-versus- k -set inclusion matrix are classical [Got66, Wil90]. Those results imply that some square full-rank minor exists. Theorem 4.18 identifies a specific base/single-swap/two-double-star minor and proves it nonsingular by direct coefficient elimination. The incidence-rank fact itself is not a novelty claim; candidate new content is the explicit sparse minor and its universal adjacent-two-value communication application.

The entropy estimate used here is elementary. It is compatible with the standard discrete entropy–variance inequality obtained by adding independent uniform noise and applying Gaussian maximal entropy [GK24].

Targeted searches found no prior theorem with the product hypothesis (16) and relative bound (33), nor a dimension-free rectangle theorem parameterized by Boolean tube radius and integer-generator entropy. This is not a substitute for an independent publication-level novelty audit.

13 Limitations and open problems

The spectral–lattice hypotheses are sufficient, not necessary. An irrational low-mode space can have useful Boolean structure not captured by integer generators. Conversely, a rational approximation may have poor Boolean margin or prohibitively large height. The robust theorem isolates the remaining inverse task: produce a projector-close rational space with simultaneously controlled Boolean margin and arithmetic entropy, or extract a coordinate/equality/code block directly.

The product-Johnson theorem has a full logarithmic one-way protocol because its rows admit the exact coordinate/complement classification. In contrast, the general spectral–lattice theorem is presently only a one-rectangle theorem: arbitrary recursive restrictions need not preserve its covariance, tube-radius, or generator-entropy hypotheses.

The most valuable next directions are:

1. reduce the polynomial depth condition $144s^4$ toward the optimal $2s$ junta threshold while retaining explicit dependence on s ;
2. unequal and near-boundary product layers with sharp tube profiles;
3. random and pseudorandom thinning inside each block;
4. a relative inverse theorem producing the rational data required by Theorem 11.1;
5. an external comparison with association-scheme and extremal product theorems not indexed by the terminology used here.

14 Machine verification

The accompanying reviewer repository contains a curated 18-module Lean 4.22 core, reduced from the broader historical research development. It contains no `sorry`, `admit`, project-defined axioms, compiled objects, or cached dependencies. The retained modules formalize the finite cores needed for the headline results:

- two-point variance and the explicit rank-at-most- $d + 1$ factorization;
- unisolvent restriction, affine support pairing, and rank-basis support pooling;

- uniform-layer and product-Johnson range/kernel lemmas;
- the short-side, low-density, and global six-coordinate sharp $2/5$ polynomials, including the unique $J(6, 3)$ equality and the $k = 2$ boundary;
- the deterministic covariance-gap medium-row contradiction and the Sylvester–Hadamard threshold.

The probabilistic sampling inputs (matrix Chernoff and VC), the projection DPP, published junta and query theorems, and the complete protocol wrapper remain conventional paper proofs. The exact coverage boundary is recorded in `formal/FORMALIZATION_MANIFEST.md`.

Five assertion-based Python suites provide independent finite regression. The current checks include 2,533,728 product-Johnson row–column pairs, 5,400 general product protocol cases, 1,000 product-multislice cases, 105,747 slice support-intersection instances, 16,952 symmetric-tree instances, 155,155 hypergeometric parameter triples, 2,047,045 adjacent intervals, 123,750 six-coordinate certificate parameters, 9,241,290 affine evaluations, and six Sylvester–Hadamard designs. The sharp audit finds the unique normalized $2/5$ equality $(n, k, |A|, a) = (6, 3, 3, 1)$.

The repository provides one-command targets for the Lean build, all Python checks, PDF generation, and a minimal arXiv source archive. Historical Fourier, affine-section, extraction, and exploratory modules are deliberately omitted because they are not dependencies of the reviewer-facing claims.

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